Classical Field Theory — Exercise Sheet 1

Local actions

Hints and answers

Use Lecture 1's sign $E_a = \partial L/\partial\phi^a - \partial_\mu p_a^\mu$. All fields and prescribed sources in this sheet are smooth. The first three problems use only differentiation, integration by parts and the fundamental lemma. The final problem also uses finite-dimensional linear ODEs.

Exercise 1.1 (A complex source and two real variations). On $\mathbb{R}^{1,1}$ let $\Phi = \phi_1 + \mathrm{i}\phi_2$ be a complex field, $J = J_1 + \mathrm{i}J_2$ a prescribed complex source, and $m \geq 0$. The density is

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\Phi_x|^2 - m^2|\Phi|^2\big) + \mathrm{Re}(\bar{J}\Phi).$$

For the action, take any compact region $D \subset \mathbb{R}^{1,1}$ and variations H whose support lies in the interior of D.

(a) Compute both real momentum currents and Euler expressions. Express the stationarity condition as one complex equation and derive the first variation for a compactly supported complex variation H.

(b) Find the source for which $\Phi(t,x) = (t^2+1) + \mathrm{i}(t^3 - x^2)$ is stationary.

(c) If only real-valued variations H are tested, must the complex equation hold? Give a counterexample with $m = 0$ and $J = 0$.

Hint. Expand the real part of $\bar{J}\Phi$ before differentiating. For compact variations, use Lecture 1 separately for ϕ_1 and ϕ_2. Real and imaginary tests detect different Euler expressions.

Answer. The equation is $(\Box + m^2)\Phi = J$. The source in the second part is $2 + m^2(t^2+1) + \mathrm{i}\big(6t + 2 + m^2(t^3 - x^2)\big)$. Real tests alone fail, for example at $\Phi = \mathrm{i}t^2$.

Exercise 1.2 (Natural conditions at the two spatial ends). Let $T, \ell > 0$, $m \geq 0$ and $\beta_0, \beta_\ell \geq 0$. On the rectangle $D = [0,T] \times [0,\ell]$, a real field has action

$$S_D[\phi] = \int_0^T \int_0^\ell \left(\frac{1}{2}\phi_t^2 - \frac{1}{2}\phi_x^2 - \frac{1}{2}m^2\phi^2 + J\phi\right) \mathrm{d}x\mathrm{d}t \\ - \frac{1}{2}\int_0^T \big(\beta_0\phi(t,0)^2 + \beta_\ell\phi(t,\ell)^2\big)\mathrm{d}t,$$

where J is prescribed. Variations vanish near $t = 0, T$ and are unrestricted at $x = 0, \ell$.

(a) Integrate coordinate by coordinate and prove that stationarity is equivalent to the bulk equation and two Robin boundary conditions. Include the case $\beta_0 = \beta_\ell = 0$.

(b) For $\ell = 1$, $m = 0$, $J = 2$ and $\beta_0 = \beta_\ell = 1$, find the unique time-independent stationary field.

Hint. The spatial integration gives $[-\phi_x H]_{x=0}^{x=\ell}$. First use interior tests, then choose independent traces at the two ends. The static sourced equation is an elementary second-order ODE.

Answer. The conditions are $\phi_{tt} - \phi_{xx} + m^2\phi = J$, $\phi_x(t,0) = \beta_0\phi(t,0)$ and $\phi_x(t,\ell) = -\beta_\ell\phi(t,\ell)$. The static field in the second part is $1 + x - x^2$.

Exercise 1.3 (Lorentz changes of coordinates). The Lorentz group in two coordinates is the matrix group

$$O(1,1) = \{\Lambda \in GL(2,\mathbb{R}) : \Lambda^{\mathsf{T}} \eta \Lambda = \eta\}, \qquad \eta = \operatorname{diag}(1,-1).$$

For $|v| < 1$, set

$$\gamma_v = (1-v^2)^{-1/2}, \qquad \begin{pmatrix} t' \\ x' \end{pmatrix} = \Lambda_v \begin{pmatrix} t \\ x \end{pmatrix}, \qquad \Lambda_v = \gamma_v \begin{pmatrix} 1 & -v \\ -v & 1 \end{pmatrix}.$$

(a) Verify that $\Lambda_v \in O(1,1)$ and $\det \Lambda_v = 1$.

(b) For a scalar field, set $\phi'(t',x') = \phi(t,x)$ at corresponding points. Use the chain rule to prove invariance of $\partial_t^2 - \partial_x^2$ and of $\frac{1}{2}(\phi_t^2 - \phi_x^2) - V(\phi)$, where V is any smooth function. Conclude equality of the scalar actions on corresponding compact regions.

Hint. The inverse boost has velocity $-v$. Express both primed derivatives in unprimed derivatives before subtracting the squares. The action also uses the coordinate Jacobian.

Answer. The primed derivatives are $\partial_{t'} = \gamma_v(\partial_t + v\partial_x)$ and $\partial_{x'} = \gamma_v(v\partial_t + \partial_x)$. Their squared difference and the scalar kinetic difference are unchanged; the Jacobian is one.

Exercise 1.4 ⋆. Let $N \geq 3$, $h_{\mathrm{lat}} = 2\pi/N$, and let $q_j(t)$, $j = 0, \dots, N-1$, be real coordinates with periodic indexing. The chain Lagrangian is

$$L_h = h_{\mathrm{lat}} \sum_{j=0}^{N-1} \left[\frac{1}{2}\dot{q}_j^2 - \frac{1}{2h_{\mathrm{lat}}^2}(q_{j+1} - q_j)^2 - \frac{1}{2}m^2 q_j^2 \right], \qquad m \geq 0.$$

(a) Derive its equations by varying every q_j.

(b) Let $T > 0$ and let ϕ be a given smooth periodic solution of $\phi_{tt} - \phi_{xx} + m^2\phi = 0$ on $[0,T] \times \mathbb{T}$. Assume $M_4 \geq 0$ and $|\partial_x^4 \phi| \leq M_4$ there. Prove that sampling this field in the chain equation produces a residual bounded by $h_{\mathrm{lat}}^2 M_4/12$ at every node.

(c) Give the chain the exact sampled initial position and velocity. Set $e_j = q_j - \phi(t, jh_{\mathrm{lat}})$, $D_h e_j = (e_{j+1} - e_j)/h_{\mathrm{lat}}$ and $\|w\|_h^2 = h_{\mathrm{lat}} \sum_j w_j^2$. With $A_h = \sqrt{2\pi} M_4 h_{\mathrm{lat}}^2/12$, prove, for $0 \leq t \leq T$,

$$\left(\|\dot{e}(t)\|_h^2 + \|D_h e(t)\|_h^2 + m^2 \|e(t)\|_h^2 \right)^{1/2} \leq A_h t, \qquad \|e(t)\|_h \leq \frac{1}{2} A_h t^2.$$

The finite linear chain has a global solution by the linear ODE theorem. No existence theorem for the continuum equation is assumed beyond the given ϕ.

Hint. Each spring contributes to two coordinate variations. For consistency, add Taylor formulas at $x + h_{\mathrm{lat}}$ and $x - h_{\mathrm{lat}}$, with fourth-derivative remainder. For convergence, differentiate the discrete quadratic error energy and use periodic summation by parts. Regularise its square root at zero.

Answer. The equations are $\ddot{q}_j - (q_{j+1} - 2q_j + q_{j-1})/h_{\mathrm{lat}}^2 + m^2 q_j = 0$. The stated estimates give order h_{lat}^2 convergence in the sampled norms on every fixed time interval, including $m = 0$.

Classical Field Theory — Exercise Sheet 2

Noether currents and their limits

Hints and answers

Use Lecture 2's signs $j^\mu = p_a^\mu Q^a - B^\mu$ and $\partial_\mu j^\mu = -E_a Q^a$. All fields and coefficients below are smooth. The unindexed $Q(t)$ denotes an integrated charge, and Q^a denotes a characteristic. The last problem is a different field theory from Klein–Gordon.

Exercise 2.1 (An improvement can change a boundary charge). Let j^μ be a current in $1+d$ dimensions and let $K^{\mu\nu} = -K^{\nu\mu}$ be smooth. Define $\widetilde{j}^\mu = j^\mu + \partial_\nu K^{\mu\nu}$. An added current with identically zero divergence is called an *identically conserved improvement* in this problem.

(a) Prove that the two currents have the same divergence for every field. Show that their charges agree on the spatial torus when K is periodic. On $\mathbb{R}^d$, assume the original slice charge exists and show equality when $K^{0i}(t,\cdot)$ is compactly supported.

(b) In $1+1$ dimensions take $K^{01} = t/(1+x^2)$ and $K^{10} = -K^{01}$. Compute the added density and flux. Find the change in total charge on $\mathbb{R}$ and the change in charge on the fixed interval $[0,1]$. Does an identically conserved improvement leave every regional charge unchanged?

Hint. Pair the terms with interchanged indices in the double divergence. The density change is a spatial derivative. Integrate that derivative before taking any limit at infinity.

Answer. The divergence change is zero. The added components in the example are $-2tx/(1+x^2)^2$ and $-1/(1+x^2)$. The total charge change is zero; on $[0,1]$ it is $-t/2$.

Exercise 2.2 (The converse with an off-shell identity supplied). For a first-order density, suppose a legitimate characteristic Q^a and specified local current j^μ satisfy $\partial_\mu j^\mu = -E_a Q^a$ for arbitrary fields, before imposing their equations. Prove that $B^\mu = p_a^\mu Q^a - j^\mu$ gives $\delta_Q L = \partial_\mu B^\mu$.

Apply this calculation on $\mathbb{R}^{1,1}$ to $L = \frac{1}{2}(\phi_t^2 - \phi_x^2)$, $\eta(t,x) = t + x$, and

$$j^0 = \eta\phi_t - \phi, \qquad j^1 = -\eta\phi_x + \phi.$$

Find its characteristic, the divergence term and an actual one-parameter point transformation realising that characteristic. Verify the off-shell identity directly.

For context, the following broader correspondence is a *cited input*, not a task to prove: in the local finite-derivative formulation for the scalar wave Euler–Lagrange equation, conservation-law classes have generalized variational characteristics. Currents are identified modulo currents vanishing on solutions and identically conserved improvements; characteristics are identified modulo ones vanishing on solutions. Here local means depending smoothly on the coordinates, field and finitely many of its derivatives. This is the scalar wave case of Olver, §5.3, Theorem 5.58, p. 334. Explain why the assertion that an arbitrary current is conserved on solutions is not the supplied off-shell identity used in the first calculation.

Hint. Start with the pointwise first-variation identity of Lecture 1. For the example, calculate the divergence before using $\phi_{tt} = \phi_{xx}$. A field shift by a prescribed function can depend on the spacetime point.

Answer. The example has $Q^1 = \eta$, $B^0 = \phi$ and $B^1 = -\phi$. The transformation is $\phi_\varepsilon(t,x) = \phi(t,x) + \varepsilon(t+x)$. Its current has divergence $\eta(\phi_{tt} - \phi_{xx}) = -E\eta$.

Exercise 2.3 (A Noether charge escapes to infinity). Let p and q be real fields on $\mathbb{R}_t \times \mathbb{R}_x$, with density $L = p(q_t + x^2 q_x)$. Choose $p_0 \in C_c^\infty((1,2))$, nonnegative and with integral one, and extend it by zero to $\mathbb{R}$. Set $q = 0$ and

$$p(t,x) = \begin{cases} (1+tx)^{-2} p_0\big(x/(1+tx)\big), & 1+tx > 0, \\ 0, & 1+tx \le 0. \end{cases}$$

(a) Derive the two field equations. Find the Noether current for the constant shift of q. Prove that the displayed fields are smooth at every finite spacetime point and satisfy both equations.

(b) Compute $Q(t) = \int_{\mathbb{R}} p(t,x)\mathrm{d}x$. Find its derivative for $t > 0$ and the limiting flux as $x \to +\infty$. Show that the initial charge is one but the charge is zero for $t \ge 1$.

(c) Identify exactly which full-space charge hypothesis in Lecture 2 fails. Why is this neither an off-shell example nor a claim about Klein–Gordon propagation?

Hint. For the transport equation use $x = \xi/(1 - t\xi)$ and retain its Jacobian. Near a point where $1 + tx = 0$, the profile argument has large absolute value, so the compact support of p_0 removes the apparent singularity. Do not differentiate the full-space integral before computing it.

Answer. The equations are $q_t + x^2 q_x = 0$ and $p_t + \partial_x(x^2 p) = 0$. The shift current is $(p, x^2 p)$. For $t \le 0$, $Q = 1$; for $t > 0$, $Q = \int_0^{1/t} p_0(\xi)\mathrm{d}\xi$. Its derivative is $-p_0(1/t)/t^2$, the negative outgoing flux. Where that flux is positive, the spatial current is not in L^1.

Classical Field Theory — Exercise Sheet 3

Stress, improvements and angular momentum

Hints and answers

Use $T^\mu{}_\nu = p^\mu \partial_\nu \phi - \delta^\mu_\nu L$, $\eta = \mathrm{diag}(1, -1, \dots, -1)$ and the inverse-metric Hilbert convention of Lecture 3. All quantities below are smooth. The first problem has d spatial coordinates; the remaining problems have three. Problems 1–3 use Lectures 1–3. Problem 4 is a return problem after Lecture 9; its electromagnetic component conventions are supplied in the question.

Exercise 3.1 (Work by an external coupling). On $I \times \mathbb{T}^d$ consider

$$L = \frac{1}{2}\phi_t^2 - \frac{1}{2}|\nabla\phi|^2 - \frac{\lambda(t)}{4}\phi^4,$$

where I is an open interval and λ is prescribed and smooth.

(a) Derive the field equation and energy density. For a classical solution, find the derivative of its total energy, including the term due to explicit time dependence. Decide whether its physical spatial momentum $P^i = \int_{\mathbb{T}^d} T^{0i} \mathrm{d}^d x$ remains constant.

(b) Set $I = (0, \pi/2)$, $\lambda(t) = 1/\sin^2 t$ and $\phi(t, x) = \sin t$. Verify the field equation, compute its energy and check the sign and value of the energy derivative by both methods.

Hint. Keep the explicit coordinate derivative of the density in the general stress identity. Spatial periodicity removes the flux, but does not remove work done by a prescribed time-dependent coupling.

Answer. The equation is $\phi_{tt} - \Delta\phi + \lambda\phi^3 = 0$ and $E' = \lambda' \int \phi^4 \mathrm{d}^d x/4$, while every P^i is constant. In the example, $E = (2\pi)^d(\frac{1}{2}\cos^2 t + \frac{1}{4}\sin^2 t)$ and $E' = -(2\pi)^d \sin t \cos t/2 < 0$.

Exercise 3.2 (Angular momentum and weighted decay). For the autonomous scalar density $L = \frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$ on $\mathbb{R}^{1+3}$, raise the second stress index with η. For fixed distinct spatial indices i, j, define

$$M^{\mu ij} = x^i T^{\mu j} - x^j T^{\mu i}.$$

(a) Prove its local conservation on solutions, keeping the derivatives of the explicit coordinate factors.

(b) Prove that $J^{ij}(t) = \int_{\mathbb{R}^3} M^{0ij} \mathrm{d}^3 x$ is constant if, on every compact time interval, every component of $T^{\mu\nu}$ is bounded by $C(1 + |x|)^{-4-\sigma}$ for some $C, \sigma > 0$. Explain why an unweighted integrability statement for stress alone does not automatically establish the angular charge hypotheses.

Hint. The coordinate derivatives leave $T^{ij} - T^{ji}$. Apply Lecture 2's charge theorem to the current $M^{\mu ij}$, which carries one additional power of $|x|$.

Answer. The local divergence vanishes because the raised scalar stress is symmetric. The stated decay gives $|M^{\mu ij}| \le 2C(1 + |x|)^{-3-\sigma}$, so the density and compact-time spatial flux satisfy Lecture 2's integrability hypotheses.

Exercise 3.3 (Belinfante improvement from a spin identity). Suppose a smooth tensor $T^{\mu\nu}$ on $\mathbb{R}^{1+3}$ is divergence-free, and a supplied smooth tensor $S^{\lambda\mu\nu}$ satisfies

$$S^{\lambda\mu\nu} = -S^{\lambda\nu\mu}, \qquad \partial_\lambda S^{\lambda\mu\nu} = T^{\nu\mu} - T^{\mu\nu}.$$

This spin identity is a given hypothesis; it is not inferred from scalar Lecture 3. Define

$$B^{\lambda\mu\nu} = \frac{1}{2}\big(S^{\lambda\mu\nu} + S^{\mu\nu\lambda} + S^{\nu\mu\lambda}\big), \qquad \Theta^{\mu\nu} = T^{\mu\nu} + \partial_\lambda B^{\lambda\mu\nu}.$$

Prove that Θ is symmetric and divergence-free. Whenever the slice integrals exist, find the difference between its integrated four-momentum and that of T. Give a sufficient boundary condition for equality, and explain why local improvement alone is insufficient.

Hint. Check $B^{\lambda\mu\nu} = -B^{\mu\lambda\nu}$ first, and then compute $B^{\lambda\mu\nu} - B^{\lambda\nu\mu}$. For the charge, the time-derivative term has $B^{00\nu} = 0$.

Answer. The two algebraic identities are $B^{\lambda\mu\nu} = -B^{\mu\lambda\nu}$ and $B^{\lambda\mu\nu} - B^{\lambda\nu\mu} = S^{\lambda\mu\nu}$. The momentum difference is the spatial boundary integral of $B^{i0\nu}$. Compact spatial support, or periodic cancellation on a torus, makes it zero.

Exercise 3.4 (Maxwell Hilbert stress, after Lecture 9). Return to this problem after Lecture 9. Let $F_{\mu\nu} = -F_{\nu\mu}$ be held fixed with both indices down while a Lorentz metric is varied. Use the density and Hilbert convention

$$L = -\frac{1}{4} g^{\mu\rho} g^{\nu\sigma} F_{\mu\nu} F_{\rho\sigma}, \qquad \delta S = \frac{1}{2}\int T_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g, \quad k^{\mu\nu} = \delta g^{\mu\nu}.$$

The volume variation is $\delta \mathrm{vol}_g = -\frac{1}{2} g_{\mu\nu} k^{\mu\nu} \mathrm{vol}_g$. Derive $T_{\mu\nu}$ by varying both inverse metrics. At $g = \eta$, set $F_{i0} = E_i$ and $F_{ij} = \epsilon_{ijk} B_k$, with $\epsilon_{123} = 1$. Compute T_{00} and T^{i0} in terms of the vectors $\mathbf{E}$ and $\mathbf{B}$. Evaluate them for $\mathbf{E} = (1, 2, 0)$ and $\mathbf{B} = (0, 0, 3)$.

Hint. Antisymmetry of F makes the two varied inverse metrics contribute equally. A lower spatial index introduces a minus sign when raised; distinguish T_{i0} from T^{i0}.

Answer. $T_{\mu\nu} = -F_{\mu\alpha} F_\nu{}^\alpha + \frac{1}{4} g_{\mu\nu} F_{\alpha\beta} F^{\alpha\beta}$. The energy density is $(|\mathbf{E}|^2 + |\mathbf{B}|^2)/2$, and $T^{i0} = (\mathbf{E} \times \mathbf{B})_i$. The example has energy density 7 and flux $(6, -3, 0)$.

Classical Field Theory — Exercise Sheet 4

Formula, stability and weak derivatives

Hints and answers

Use the unitary Fourier transform $\widehat{f}(k) = (2\pi)^{-1/2}\int_{\mathbb{R}} e^{-ikx}f(x)\mathrm{d}x$ and $\|f\|_{H^1}^2 = \int_{\mathbb{R}}(1+k^2)|\widehat{f}(k)|^2\mathrm{d}k$. All wave equations on this sheet have one spatial dimension and zero mass. The solution class is that of Theorem 4.3.

Exercise 4.1 (Two initial data and two travelling packets). Let $f, g \in C_c^\infty(\mathbb{R};\mathbb{R})$ be initial displacement and velocity.

(a) Derive d'Alembert's formula from Lecture 4's Fourier multipliers. Use an oriented integral when $t < 0$. Verify the equation and both data, and justify identifying the formula with the energy-class solution.

(b) Let F, G be smooth real functions supported in $[-1/4, 1/4]$ and strictly positive on its interior. Take

$$f(x) = F(x+3) + 2G(x-2), \qquad g(x) = -F'(x+3) + 2G'(x-2).$$

Find the solution. For $t \geq 0$, find the open time intervals on which an observer at $x = 0$ sees each contribution. Where and when do the packet centres meet? Find the corresponding visibility interval at that position.

Hint. Write $\cos(tk)$ as two exponentials and $S_t(|k|) = \int_0^t \cos(sk)\mathrm{d}s$. Translation in physical space then turns the sine term into an integral of g.

Answer. $v = \frac{1}{2}[f(x+t)+f(x-t)] + \frac{1}{2}\int_{x-t}^{x+t} g$. The packets are $F(x+3-t)+2G(x-2+t)$; at zero their intervals are $(11/4, 13/4)$ and $(7/4, 9/4)$. Their centres meet at $(t, x) = (5/2, -1/2)$; both are visible there for $t \in (9/4, 11/4)$.

Exercise 4.2 (Hadamard instability for an elliptic Cauchy problem). For each positive integer n, seek a smooth function $u_n(x, y)$, periodic in $x \in \mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, satisfying

$$\partial_y^2 u_n + \partial_x^2 u_n = 0, \qquad u_n(x, 0) = 0, \qquad \partial_y u_n(x, 0) = e^{-\sqrt{n}}\cos(nx).$$

Find such a solution by separation of variables. Prove that the initial velocity tends to zero in every fixed $C^r(\mathbb{T})$ seminorm, while $\|u_n(\cdot, y)\|_{L^2(\mathbb{T})} \to \infty$ for every fixed $y > 0$. What continuity claim does this disprove? Why does it not contradict Lecture 4?

Hint. The amplitude solves $a_n'' - n^2 a_n = 0$. Compare the logarithms of $n^r e^{-\sqrt{n}}$ and $e^{-\sqrt{n}}\sinh(ny)/n$ as $n \to \infty$.

Answer. $u_n = e^{-\sqrt{n}}\sinh(ny)\cos(nx)/n$ and $\|u_n(\cdot, y)\|_2 = \sqrt{\pi}e^{-\sqrt{n}}\sinh(ny)/n$. The elliptic Cauchy solution map from smooth data to the slice L^2 norm is discontinuous at zero; the equation has the opposite spatial sign to the wave equation.

Exercise 4.3 (A corner and its weak derivatives ⋆). Let $f(x) = e^{-|x|}$. Compute its unitary Fourier transform and its H^1 norm. Identify its first weak derivative by testing against every $\zeta \in C_c^\infty(\mathbb{R})$, and calculate its second distributional derivative. Decide whether f belongs to the space

$$H^2(\mathbb{R}) = \left\{ h \in L^2(\mathbb{R}) : \int_{\mathbb{R}} (1+k^2)^2 |\widehat{h}(k)|^2 \mathrm{d}k < \infty \right\}.$$

Distinguish the value assigned to a weak derivative at the single point zero from the existence of a classical derivative there.

Hint. Split each integral at zero. The endpoint terms cancel for the first derivative and add for the derivative of the first derivative.

Answer. $\widehat{f} = \sqrt{2/\pi}/(1+k^2)$, $\|f\|_{H^1}^2 = 2$, $f' = -\operatorname{sgn}(x) e^{-|x|}$ almost everywhere, and $f'' = f - 2\delta_0$. Thus $f \in H^1 \setminus H^2$ and has no classical derivative at zero.

Exercise 4.4 (The sharp massless sine bound). Choose a real even $\rho \in C_c^\infty((-1,1))$ with $\|\rho\|_2 = 1$. For $0 < \varepsilon \le 1$, let g_ε satisfy $\widehat{g}_\varepsilon(k) = \varepsilon^{-1/2}\rho(k/\varepsilon)$, and let v_ε have initial data $(0, g_\varepsilon)$.

(a) Prove that g_ε is real and normalized in L^2, and that for every $T > 0$,

$$\sup_{|t| \le T} \|v_\varepsilon(t) - t g_\varepsilon\|_2 \le \frac{T^3 \varepsilon^2}{6}.$$

(b) For fixed real t, determine the operator norm on real $L^2(\mathbb{R})$ of the Fourier multiplier $S_t(|k|)$, where $S_t(0) = t$. Explain why conservation of massless energy cannot give a bound for $\|v(t)\|_2$ uniform in both time and all initial data with energy $1/2$.

Hint. Integrate the estimate $|1 - \cos s| \le s^2/2$ to control $\sin z - z$. The shrinking Fourier support makes the sine multiplier approach its zero-frequency value uniformly on each bounded time interval.

Answer. $\|g_\varepsilon\|_2 = 1$ and the stated error bound holds. The multiplier norm is exactly $|t|$, including zero at $t = 0$. All these data have massless energy $1/2$, but their displacement norms approach $|t|$ for every fixed t as $\varepsilon \to 0$.

Classical Field Theory — Exercise Sheet 5

Cones with forcing and interior tails

Hints and answers

Write $\Box = \partial_t^2 - \Delta$. Every cone argument below concerns real C^2 fields on an open neighbourhood of the closed cone $\{0 \le t \le R,\ |x-a| \le R-t\}$, with $R > 0$. No existence theorem for a nonlinear equation is assumed or proved here.

Exercise 5.1 (Source work on a shrinking cone). Suppose $(\Box + m^2)w = F$, with $m \ge 0$ and continuous F. Let $E_m(t)$ be the integral over $B_{R-t}(a)$ of $e_m = (w_t^2 + |\nabla w|^2 + m^2 w^2)/2$.

(a) Derive the exact shrinking-ball balance, including source work and all lateral flux terms. Prove, for $0 \le t < R$,

$$E_m(t) \le e^t \left[E_m(0) + \frac{1}{2} \int_0^t e^{-s} \|F(s)\|^2_{L^2(B_{R-s}(a))} \mathrm{d}s \right].$$

(b) If $F = 0$ on the cone and both initial data vanish on its base, prove that $w = 0$ on the closed cone, including when $m = 0$. Explain why this is a local statement even if total energy is infinite.

Hint. The local balance has source term $w_t F$. Retain the inward motion of the ball, estimate source work by two squares, and multiply the resulting differential inequality by e^{-t}.

Answer. $E_m' = \int_{B_{R-t}} w_t F - \frac{1}{2} \int_{\partial B_{R-t}} [(w_t - \partial_n w)^2 + |\nabla_{\tan} w|^2 + m^2 w^2]$. Thus $E_m' \le E_m + \|F\|_2^2/2$, giving the bound. Zero data and source give zero field by integrating its vanishing velocity along vertical segments.

Exercise 5.2 (Finite speed for a semilinear equation, conditional on solutions). Let $V \in C^2(\mathbb{R})$ and suppose two real C^2 fields ϕ_1, ϕ_2 solve $\Box\phi + V'(\phi) = 0$ on a neighbourhood of the same closed cone. Assume their displacements and velocities agree on $B_R(a)$ at time zero.

(a) Prove that the fields agree on the closed cone. Use the mass-one energy of their difference, allowing V'' to have either sign. Give an explicit finite coefficient in the energy inequality in terms of a bound for $|V''|$ on the field values occurring in the cone.

(b) Suppose instead that the fields are defined on an open slab $I \times \mathbb{R}^d$, with $0 \in I$, and their two initial data agree outside a closed set K. Prove agreement at every (t, x) with $\mathrm{dist}(x, K) > |t|$. What extra condition makes the zero field an available comparison solution?

Hint. Write $V'(\phi_1) - V'(\phi_2) = b(\phi_1 - \phi_2)$ by integrating V'' along the segment between the two field values. In the equation for the difference, put $+w$ on the left and $(1-b)w$ on the right.

Answer. If L bounds $|V''|$ on the compact field range, the difference's mass-one cone energy satisfies $E_1' \le (1+L)E_1$. Zero initial energy implies agreement. Compact perturbations stay in $\mathrm{dist}(x, K) \le |t|$; comparison with zero requires $V'(0) = 0$.

Exercise 5.3 (Huygens' principle and the dimension). Consider the massless wave with zero displacement and real $g \in C_c^\infty(\mathbb{R}^d)$ as initial velocity. The following classical representation formulas may be used as given facts for $t > 0$:

$$u_1(t,x) = \frac{1}{2}\int_{x-t}^{x+t} g(y)\mathrm{d}y, \qquad u_2(t,x) = \frac{1}{2\pi}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\mathrm{d}^2y,$$

$$u_3(t,x) = \frac{1}{4\pi t}\int_{|y-x|=t} g(y)\mathrm{d}S_y.$$

For the higher-dimensional classification one may use these general forms. Let α_d be the volume of the unit ball, $\fint$ the integral divided by the region's volume or area, and $D_t = t^{-1}\partial_t$. For odd $d \geq 3$ and even $d \geq 2$, respectively,

$$u_d = \frac{1}{1\cdot 3\cdots(d-2)} D_t^{(d-3)/2}\left[t^{d-2}\fint_{|y-x|=t} g(y)\mathrm{d}S_y\right],$$

$$u_d = \frac{1}{2\cdot 4\cdots d} D_t^{(d-2)/2}\left[\frac{1}{\alpha_d}\int_{|y-x|<t} \frac{g(y)}{\sqrt{t^2-|y-x|^2}}\mathrm{d}^dy\right].$$

These facts hold for the smooth compact data specified here; see Evans, *Partial Differential Equations*, 2nd ed., §2.4.1, formulas (31), (38), Theorems 2–3, pp. 73–77. Nonzero displacement contributes the time derivative of the corresponding velocity formula with g replaced by that displacement.

(a) For $d = 1, 2, 3$, let $g \geq 0$, $g \not\equiv 0$, with support inside $B_1(0)$. Decide whether $u_d(t,0)$ vanishes for $t > 1$. Explain the difference between finite speed and dependence only on data arbitrarily near the base sphere.

(b) Use the general formulas to classify all positive spatial dimensions. For even d, test data supported strictly inside the base sphere so that every time derivative acts on a smooth kernel. Explain why a statement that all even-dimensional kernels are positive would be too strong.

Hint. For the tail test the integration domain contains the fixed support of g. Repeated application of D_t to $(t^2-r^2)^{-1/2}$ has alternating sign and a nonzero coefficient. In odd dimensions a whole neighbourhood of the base sphere missing the data makes all the displayed derivatives zero.

Answer. For $t > 1$, $u_1(t,0)$ and $u_2(t,0)$ are positive, while $u_3(t,0) = 0$. Huygens' sphere property holds for odd $d \geq 3$, but fails for $d = 1$ and every even d. Higher even-dimensional interior kernels alternate in sign.

Classical Field Theory — Exercise Sheet 6

Causal kernels and frequency boundary values

Hints and answers

Use $\Box = \partial_t^2 - \Delta$ and Lecture 6's normalized kernel $G_{\mathrm{ret}} = \delta(t-|x|)/(4\pi|x|)$ in three spatial dimensions. Its meaning at the vertex is its test pairing, not a product of distributions. All distribution pairings are bilinear. Write H for the Heaviside function; its value at zero does not affect the distributions below.

Exercise 6.1 (An entire radial source response). Let $q \in C_c^\infty(\mathbb{R})$, and let h be a smooth even compactly supported function such that $x \mapsto h(|x|)$ is smooth on $\mathbb{R}^3$. Take $f(t,x) = q(t)h(|x|)$ and $u = G_{\mathrm{ret}} * f$.

(a) Perform the angular integration to obtain a one-dimensional radial integral for $u(t,r)$, $r = |x| > 0$, with an inner integral over travel distance. Find its limit at $r = 0$ and explain why no singularity remains there.

(b) Set $v(t,r) = ru(t,r)$ for $r \geq 0$ and extend it oddly to the real line. Reduce the equation for v to a forced one-dimensional wave and derive its causal double-integral formula. Check its agreement with part (a).

(c) Let $a < b$ and $R > 0$. If $\operatorname{supp} q \subset [a,b]$ and $\operatorname{supp} h \subset [-R,R]$, find the possible response interval at an observer $r > R$. If $q > 0$ on (a,b) and $h > 0$ for $|s| < R$, prove that the response is positive throughout the interior of that interval.

Hint. At fixed source radius s, use the travel distance $\ell = (r^2 + s^2 - 2rs\cos\theta)^{1/2}$ as angular variable. The one-dimensional forcing is the odd function $sh(|s|)q(t)$.

Answer. $u(t,r) = \frac{1}{2r}\int_0^\infty sh(s)\int_{|r-s|}^{r+s} q(t-\ell)\mathrm{d}\ell\mathrm{d}s$ and $u(t,0) = \int_0^\infty sh(s)q(t-s)\mathrm{d}s$. The odd field satisfies $v_{tt} - v_{rr} = rh(|r|)q(t)$. For $r > R$ the response is supported in $[a+r-R, b+r+R]$.

Exercise 6.2 (Which side of the frequency poles?). The spacetime Fourier convention is

$$\widetilde{F}(\nu,k) = (2\pi)^{-2}\int e^{i\nu t - ik\cdot x}F(t,x)\mathrm{d}t\mathrm{d}^3x.$$

For a tempered distribution it is defined by transposition on Schwartz tests, with this same kernel. Fourier transformation preserves Schwartz space and its topology, so this definition preserves convergence of tempered distributions. The Schwartz Fourier automorphism is the one from *Analysis* I used in Lecture 4; transposition here specifies its distributional extension. Let $G_\varepsilon = e^{-\varepsilon t}G_{\mathrm{ret}}$, $\varepsilon > 0$; the product is understood on the future cone, where the factor is bounded.

(a) Compute its spatial transform, then its full spacetime transform. Prove convergence as $\varepsilon \downarrow 0$ on Schwartz tests and express $\widetilde{G}_{\mathrm{ret}}$ as a frequency boundary value. Check the Fourier coefficient of the equation $\Box G_{\mathrm{ret}} = \delta^4$.

(b) For fixed $\kappa = |k| > 0$, derive the principal-value and delta terms of that boundary value. Here $\langle \mathrm{pv}(1/s), \zeta\rangle = \lim_{a\downarrow 0}\int_{|s|>a}\zeta(s)/s\,\mathrm{d}s$. Derive the identity for $(s \pm i0)^{-1}$ by separating real and imaginary parts. Compare the advanced and half-retarded plus half-advanced kernels. State why the fixed-κ decomposition cannot simply be evaluated at $\kappa = 0$.

Hint. The spatial transform of the spherical shell uses $\int_{S^2} e^{-it k\cdot\theta}\mathrm{d}\Omega = 4\pi\,\sin(t\kappa)/(t\kappa)$. Then integrate $e^{-(\varepsilon-i\nu)t}\sin(t\kappa)/\kappa$. For the pole identity, $\varepsilon/(s^2+\varepsilon^2)$ is an approximate delta with integral π.

Answer. $\widetilde{G}_\varepsilon = (2\pi)^{-2}[|k|^2-(\nu+i\varepsilon)^2]^{-1}$. At $\kappa>0$ the boundary value inside the brackets is $\mathrm{pv}(\kappa^2-\nu^2)^{-1}+\frac{i\pi}{2\kappa}[\delta(\nu-\kappa)-\delta(\nu+\kappa)]$. Advanced reverses the imaginary term; their half-sum retains the principal value.

Exercise 6.3 (Descending the spatial dimension). Starting with the three-dimensional retarded pairing, integrate out one auxiliary spatial coordinate to construct a kernel G_2 on $\mathbb{R}^{1+2}$, and two auxiliary coordinates to construct G_1 on $\mathbb{R}^{1+1}$. Make the construction legitimate on compact smooth tests: a test independent of the auxiliary coordinates is not itself compactly supported. Derive explicit locally integrable formulas, prove $(\partial_t^2-\Delta_d)G_d=\delta^{1+d}$ for $d=1,2$, and identify their support and their relation to Sheet 5's interior tails.

Hint. Lift a compact test with an auxiliary cutoff equal to one on a ball larger than its time support. On the cone, the auxiliary radius is bounded by that time support. For two removed coordinates use polar coordinates and then the full travel distance as integration variable.

Answer. $G_2(t,x)=H(t-|x|)/(2\pi\sqrt{t^2-|x|^2})$ inside the future cone, and $G_1(t,x)=H(t-|x|)/2$. Both are retarded fundamental solutions, with support the whole closed future cone.

Classical Field Theory — Exercise Sheet 7

Modes, complex pairs and admissibility

Hints and answers

Use $\{F,G\} = \int (F_\phi G_\pi - F_\pi G_\phi)$ and $\Omega = \int \delta\phi \wedge \delta\pi$. Except for the explicitly finite torus model in Problem 1, functionals have Lecture 7's smooth Schwartz-valued gradients. A formal distributional derivative does not automatically belong to that class.

Exercise 7.1 (Real oscillators in a finite torus model). On $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$, fix an integer $N \geq 2$ and use the real basis

$$e_0 = (2\pi)^{-1/2}, \qquad e_{n,c} = \pi^{-1/2}\cos(nx), \qquad e_{n,s} = \pi^{-1/2}\sin(nx), \quad 1 \leq n \leq N.$$

Let $\phi = q_0 e_0 + \sum_{n=1}^{N}(q_{n,c}e_{n,c} + q_{n,s}e_{n,s})$ and let π have the same expansion with coefficients p.

(a) Verify orthonormality and compute the pulled-back canonical form and the Hamiltonian $H = \frac{1}{2}\int_{\mathbb{T}}(\pi^2 + \phi_x^2 + m^2\phi^2)\mathrm{d}x$, where $m \geq 0$. Derive the oscillator equations and treat the zero mode at $m = 0$.

(b) Define $\widehat{\phi}_n = (2\pi)^{-1/2}\int e^{-inx}\phi\mathrm{d}x$, and similarly $\widehat{\pi}_n$, for $|n| \leq N$. Express them in the real coordinates and compute their complex bilinear brackets, including $n = 0$.

(c) Compute all nonzero real coefficients and the energy of the data $\phi = 3\cos(2x) - 2\sin x$, $\pi = 5\sin(2x) + \cos x$.

Hint. Integrate products of sines and cosines before reading off the canonical coordinates. The complex Fourier coefficient contains both a cosine and a sine coordinate; reality relates the opposite modes.

Answer. The form is $\delta q_0 \wedge \delta p_0 + \sum_{n,\alpha=c,s}\delta q_{n,\alpha} \wedge \delta p_{n,\alpha}$. Each frequency is $\sqrt{n^2+m^2}$ and $\{\widehat{\phi}_n, \widehat{\pi}_l\} = \delta_{n+l,0}$. The given data have energy $\pi(33 + 13m^2/2)$.

Exercise 7.2 (A complex scalar is two real canonical pairs). Write $\Phi = \phi_1 + i\phi_2$, with both real fields Schwartz, and take

$$L = \frac{1}{2}\big(|\Phi_t|^2 - |\nabla\Phi|^2 - m^2|\Phi|^2\big), \qquad m \geq 0.$$

Derive the two momenta and Hamiltonian by real variation. Compute all functional derivatives of $Q = \int(\phi_1\pi_2 - \phi_2\pi_1)\mathrm{d}^d x$ and its action on both fields and both momenta. Identify its finite flow, prove $\{Q,H\} = 0$, and verify admissibility in the product of the two real Schwartz phase spaces. Evaluate Q on $(\phi_1,\phi_2;\pi_1,\pi_2) = (f, 2f; -f, 3f)$, where $f \in \mathcal{S}(\mathbb{R}^d;\mathbb{R})$ and $\|f\|_2 = 1$.

Hint. First expand the absolute squares into their two real terms. The charge must rotate the momentum components as well as the field components. For conservation, integrate the two Laplacian terms by parts.

Answer. $\pi_a = \partial_t\phi_a$ and $H = \frac{1}{2}\sum_{a=1}^{2}\int(\pi_a^2 + |\nabla\phi_a|^2 + m^2\phi_a^2)$. The flow multiplies both Φ and $\pi_1 + i\pi_2$ by $e^{i\alpha}$; $\{Q,H\} = 0$. The given data have $Q = 5$.

Exercise 7.3 (A smooth window and a sharp interval). In one dimension let $w \in C_c^\infty(\mathbb{R};\mathbb{R})$ be a fixed window and set

$$F_w[\phi,\pi] = \frac{1}{2}\int_{\mathbb{R}} w(x)\phi_x(x)^2 \mathrm{d}x, \qquad F_\#[\phi,\pi] = \frac{1}{2}\int_0^1 \phi_x(x)^2 \mathrm{d}x.$$

Compute both derivatives against arbitrary real Schwartz variations. Prove F_w is admissible. Is $F_\#$ an admissible functional on the whole Schwartz phase space? Test the field $\phi(x) = (x+x^2)\eta(x)$, where $\eta \in C_c^\infty(\mathbb{R})$ equals one on a neighbourhood of $[0,1]$. Display its endpoint distribution terms.

Hint. Do not discard the two interval endpoint terms. To rule out a smooth representing gradient, test it with bounded bumps shrinking to one endpoint.

Answer. $\operatorname{grad} F_w = (-\partial_x(w\phi_x), 0)$ is Schwartz. The sharp derivative is $-\mathbf{1}_{(0,1)}\phi_{xx} + \phi_x(1)\delta_1 - \phi_x(0)\delta_0$. For the specified field it is $-2\mathbf{1}_{(0,1)} + 3\delta_1 - \delta_0$, so $F_\#$ is not admissible on the whole phase space.

Exercise 7.4 (The sign of the Hamiltonian commutator). For admissible F, G, let $H_F = D(\operatorname{grad} F)$ and $\mathbb{J}(a,b) = (b,-a)$ as in Lecture 7. Define the vector-field bracket by

$$[X,Y](z) = DY(z)X(z) - DX(z)Y(z),$$

the sign for which it acts on functionals as $X(YK) - Y(XK)$. Use the actual bracket-gradient formula of Lecture 7 to prove $X_{\{F,G\}} = -[X_F, X_G]$. For a nonzero example, choose real $f \in \mathcal{S}$ with $\|f\|_2 = 1$, put $q = \int f\phi$, $p = \int f\pi$, and take $F = q^2/2$, $G = p^2/2$. Compute both vector fields, their commutator, and $X_{\{F,G\}}$.

Hint. Differentiate $X_F = \mathbb{J}\operatorname{grad} F$. In the example the smeared coordinates satisfy $\{q,p\} = 1$, so the commutator must be nonzero.

Answer. $\operatorname{grad}\{F,G\} = H_F\mathbb{J}\operatorname{grad} G - H_G\mathbb{J}\operatorname{grad} F$ gives the minus sign. In the example $[X_F, X_G] = (-qf, pf)$, whereas $X_{\{F,G\}} = (qf, -pf)$.

Classical Field Theory — Exercise Sheet 8

Slices, boundary conditions and causal brackets

Hints and answers

For real Klein–Gordon variations with compact smooth Cauchy data, use $\omega^\mu(h_1, h_2) = h_1\partial^\mu h_2 - h_2\partial^\mu h_1$. The future-oriented graph element is $(1, -\nabla\tau)\mathrm{d}^d x$. All fields below are smooth. The last problem explicitly restricts the field equation to the massless wave in $3+1$ dimensions.

Exercise 8.1 (An unbounded Cauchy graph). Let $\tau(x) = t_* + v \cdot x$ on $\mathbb{R}^d$, where $t_* \in \mathbb{R}$ and $v \in \mathbb{R}^d$, $|v| < 1$. Let $h_1, h_2 \in \mathcal{S}_m$ be the actual compact-Cauchy-data solutions of Lecture 8, $m \geq 0$.

(a) Prove that the graph is spacelike and meets each inextendible timelike curve exactly once. A future timelike curve can be parametrized by coordinate time with $|\dot{x}(t)| < 1$; justify that its time range is all of $\mathbb{R}$. Here an inextendible curve has no past or future endpoint in Minkowski spacetime.

(b) Although τ is unbounded, prove that the current is supported in a compact subset of the region between the graph and $t = 0$. Give an explicit spatial bound in terms of t_*, $|v|$, and a radius R containing both sets of initial data. Use signed time integration to prove $\Omega_{\Sigma_\tau}(h_1, h_2) = \Omega_0(h_1, h_2)$, including where the graph crosses zero.

(c) Choose nonzero real $f \in C_c^\infty(\mathbb{R}^d)$ and solutions with data $(f, 0)$ and $(0, f)$. Determine their pairing on this graph.

For comparison, the following general result is supplied: in full Minkowski spacetime, the same compact-data solutions have finite, equal symplectic integrals on every smooth spacelike Cauchy surface with future orientation. It uses spacelike compact support, not just current closure. See Khavkine, §3.3.3, Lemma 3.11, p. 41, arXiv:1402.1282. Your proof here should establish the affine case directly.

Hint. Along a timelike curve differentiate $t - t_* - v \cdot x(t)$. In the comparison region combine $|x| \leq R + |t|$ with $|t| \leq |t_*| + |v||x|$. Once compactness is proved, Lecture 8's signed-FTC argument applies without a compact graph deformation.

Answer. The relevant support satisfies $|x| \leq (R + |t_*|)/(1 - |v|)$. The graph pairing equals the flat one, and for the two specified solutions it is $\int f^2 \mathrm{d}^d x > 0$.

Exercise 8.2 (Common boundary conditions remove side flux). Let two real solutions of $(\partial_t^2 - \partial_x^2 + m^2)h = 0$ be smooth on a neighbourhood of $I \times [0, L]$, where I is an open interval, $L > 0$ and $m \geq 0$. Set $\Omega_{[0,L]}(h_1, h_2) = \int_0^L (h_1 h_{2,t} - h_2 h_{1,t})\mathrm{d}x$. At each endpoint, consider the following common conditions on both solutions: Dirichlet $h = 0$, Neumann $\partial_n h = 0$, or real Robin $\partial_n h + \kappa h = 0$, with the same prescribed real κ for both solutions at that endpoint. The two endpoints may have different such conditions.

(a) Prove that each choice makes the side symplectic flux zero and conserves $\Omega_{[0,L]}$. Specify the outward derivative at each endpoint.

(b) For $m = 0$, $L = \pi$, compare $h_D(t, x) = \sin x \cos t$ and $h_N(t, x) = 1$. Each obeys one of the listed conditions, but they obey different conditions. Compute their current, pairing and net outward side flux. Is individual satisfaction of some listed condition sufficient?

Hint. The outward side flux is the negative of $h_1\partial_n h_2 - h_2\partial_n h_1$. The cancellation requires the same endpoint relation for both arguments.

Answer. The common conditions give zero side flux and a constant pairing. For the mixed example, $\omega^0 = \sin x \sin t$, $\omega^1 = \cos x \cos t$, $\Omega = 2\sin t$, and the net outward flux is $-2\cos t$. Different conditions do not suffice.

Exercise 8.3 (A causal source bracket and its sign). In $3+1$ dimensions let G_{ret} be Lecture 6's massless kernel, let $G_{\mathrm{adv}}(t,x) = G_{\mathrm{ret}}(-t,x)$, and put $E = G_{\mathrm{adv}} - G_{\mathrm{ret}}$. For real $f, g \in C_c^\infty(\mathbb{R}^{1+3})$, define on the massless compact-data solution space

$$F_f(\phi) = \int_{\mathbb{R}^{1+3}} f\phi \, \mathrm{d}t\mathrm{d}^3x, \qquad Ef = (G_{\mathrm{adv}} - G_{\mathrm{ret}}) * f.$$

The identification of the resulting causal bracket with the linear *Peierls bracket* may be taken as given; see Khavkine, §§3.3.4–3.3.5, Theorem 3.2, and the scalar case §4.1, arXiv:1402.1282. The signs in this problem must be fixed by the following calculation, using the course convention $\iota_{X_F}\Omega = DF$.

(a) Show that Ef is a smooth homogeneous solution with compact smooth data on every constant slice. Apply the sourced current balance to $G_{\mathrm{ret}} * f$ and a homogeneous compact-data solution h to prove $\Omega(Ef, h) = \int fh$. Identify X_{F_f} and derive the bracket $\{F_f, F_g\} = \int f(Eg)$. Prove its antisymmetry and its vanishing when the two test supports are spacelike separated.

(b) As a separate finite-dimensional sign test, take one oscillator $\ddot{q} + \nu^2 q = 0$, $\nu \geq 0$, with $\{q(0), p(0)\} = 1$ and $p = \dot{q}$. Compute $\{q(t), q(s)\}$ and $\{q(t), p(s)\}$, using continuous limits at $\nu = 0$. Recover the equal-time canonical sign. Explain why this finite model does not make point evaluation of a field admissible in Lecture 7.

Hint. For $r_f = G_{\mathrm{ret}} * f$, the current with h has divergence $-hf$. Compare slices before and after the source; on the late slice $Ef = -r_f$. The advanced field vanishes there.

Answer. $X_{F_f} = Ef$ and $\{F_f, F_g\} = \int f(Eg)$. The oscillator brackets are $\sin(\nu(s-t))/\nu$ and $\cos(\nu(s-t))$, with limits $s - t$ and 1. In particular $\{q(t), p(t)\} = 1$.

Classical Field Theory — Exercise Sheet 9

Gauge identities and source conditions

Hints and answers

Use $A_{\mathrm{em}} = -\phi_{\mathrm{em}}\mathrm{d}t + \mathbf{A}\cdot\mathrm{d}x$, $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t\mathbf{A}$ and $\mathbf{B} = \operatorname{curl}\mathbf{A}$. Unless a weaker class is explicitly discussed, fields and coefficients are smooth. Compact support is stated separately in each problem.

Exercise 9.1 (The adjoint differentiates coefficients too). Let a smooth first-order density $L(x,\phi,\partial\phi)$ have Euler expressions E_a on an open coordinate region $U \subset \mathbb{R}^{1+d}$. Vary its action on a compact region $D \subset U$, with every parameter supported in the interior of D. For smooth parameters ϵ^α, consider the finite-order variation

$$\delta_\epsilon\phi^a = \sum_{|I|\le r} R^{aI}_\alpha(x, j^s\phi)\partial_I\epsilon^\alpha.$$

Here I is a derivative multi-index, each multi-index occurs once in the sum, and $j^s\phi$ denotes the field and all its derivatives through finite order s. The coefficients are smooth functions of these arguments. Assume the action is invariant under these variations for every compact parameter and every smooth field, without imposing the field equations.

(a) Derive the identity satisfied by the Euler expressions, including the variable coefficients. For the one-dimensional test operator $R\epsilon = (1+x^2)\epsilon_x + x\epsilon$, display R^*E explicitly. This test asks for its adjoint, not for a claim that every density has that symmetry.

(b) For Maxwell's transformation $\delta\phi_{\mathrm{em}} = -\epsilon_t$, $\delta\mathbf{A} = \nabla\epsilon$, derive the identity for the scalar and vector Euler expressions. With $\mathcal{M}_0 = \operatorname{div}\mathbf{E}$ and $\boldsymbol{\mathcal{M}} = \partial_t\mathbf{E} - \operatorname{curl}\mathbf{B}$, include a prescribed source $(\rho,\mathbf{J})$ and identify its necessary conservation condition. Explain the distributional interpretation for only C^2 potentials.

Hint. Apply the compact first-variation formula before moving derivatives off the parameter. Each integration by parts differentiates the full product $R^{aI}_\alpha E_a$.

Answer. $\sum_{a,I}(-1)^{|I|}\partial_I(R^{aI}_\alpha E_a) = 0$. The test operator gives $R^*E = -(1+x^2)E_x - xE$. For Maxwell, $\partial_t E_\phi - \operatorname{div}E_A = -(\rho_t + \operatorname{div}\mathbf{J})$, so invariance requires a conserved source.

Exercise 9.2 (Polarization and magnetization). Let $P, M \in C^\infty_c(\mathbb{R}^{1+3};\mathbb{R}^3)$ and define

$$\rho = -\operatorname{div}P, \qquad \mathbf{J} = \partial_t P + \operatorname{curl}M.$$

Prove conservation, zero total charge on each slice and $\int y\rho(t,y)\mathrm{d}^3y = \int P(t,y)\mathrm{d}^3y$. For any smooth potential, rewrite the integrated source coupling $\int(-\rho\phi_{\mathrm{em}} + \mathbf{J}\cdot\mathbf{A})\mathrm{vol}$ in terms of P, M and the fields, displaying why all discarded terms vanish. Does this construction assert that the potential of this source is unique?

Hint. Use the divergence of a curl, then coordinate integration by parts in space and time. The curl term transfers to $\mathbf{A}$ with a plus sign.

Answer. The source is conserved and neutral, its dipole is $\int P$, and its coupling is $\int(P\cdot\mathbf{E} + M\cdot\mathbf{B})\mathrm{vol}$. Gauge-related potentials are still not unique.

Exercise 9.3 (Vacuum duality with the Lorentzian star). For a vacuum Maxwell field, put $F' = \star F_{\mathrm{em}}$, using exactly the Hodge table of Lecture 9 and $\star^2 = -1$ on two-forms. Find the transformed electric and magnetic vectors, and verify all four vacuum component equations. Determine the effect of two and four successive transformations. Check the energy and Poynting densities for the constant vacuum fields $\mathbf{E} = (1, 0, 2)$, $\mathbf{B} = (0, 3, 0)$ before and after the map. These are densities, not finite total energies on $\mathbb{R}^3$. Does the same transformation preserve Maxwell's equations with a prescribed nonzero electric current three-form $\mathcal{J}$ and no magnetic source?

Hint. Express the starred two-form again in the basis $\mathrm{d}x^i \wedge \mathrm{d}t$ and the spatial magnetic wedges. Their order matters.

Answer. $(\mathbf{E}, \mathbf{B}) \mapsto (-\mathbf{B}, \mathbf{E})$; twice gives $(-\mathbf{E}, -\mathbf{B})$ and four times the identity. Both examples have energy density 7 and flux $(-6, 0, 3)$. For nonzero electric source, $\mathrm{d}F' = \mathcal{J}$ rather than zero.

Exercise 9.4 (Conservation does not remove a time-face term). Let $D = [t_0, t_1] \times \mathbb{T}^3$, $t_1 > t_0$, and use the Maxwell density $L = \frac{1}{2}(|\mathbf{E}|^2 - |\mathbf{B}|^2) - \rho\phi_{\mathrm{em}} + \mathbf{J} \cdot \mathbf{A}$. Take smooth periodic fields and a prescribed smooth conserved source. Allow a smooth periodic gauge function $\chi(t, x)$ with nonzero time-face values.

(a) Find the exact action change, retaining the time faces and explaining the cancellation on opposite spatial faces.

(b) Set $\rho = \sin x^1$, $\mathbf{J} = 0$, $\phi_{\mathrm{em}} = \sin x^1$, $\mathbf{A} = 0$, and $\chi = \gamma t \sin x^1$, where real $\gamma \neq 0$. Verify Maxwell's equations before and after the transformation and compute the nonzero action change. Why is this consistent with Lecture 9's compact-parameter gauge invariance?

Hint. The kinetic density is unchanged. Rewrite $\rho\chi_t + \mathbf{J} \cdot \nabla\chi$ as a spacetime divergence minus $\chi(\rho_t + \operatorname{div} \mathbf{J})$.

Answer. $S[A_{\mathrm{em}} + \mathrm{d}\chi] - S[A_{\mathrm{em}}] = [\int_{\mathbb{T}^3} \rho\chi \mathrm{d}^3x]_{t_0}^{t_1}$. The example has unchanged fields $\mathbf{E} = (-\cos x^1, 0, 0)$, $\mathbf{B} = 0$ and action change $4\pi^3\gamma(t_1 - t_0)$.

Classical Field Theory — Exercise Sheet 10

Residual gauges and charged tails

Hints and answers

Write $\Box = \partial_t^2 - \Delta$. Use $\mathbf{E} = -\nabla\phi_{\mathrm{em}} - \partial_t \mathbf{A}$, $\mathbf{B} = \operatorname{curl}\mathbf{A}$ and $\ell = -(\partial_t\phi_{\mathrm{em}} + \operatorname{div}\mathbf{A})$. Lecture 10's theorem assumes a smooth conserved source of compact *spacetime* support and compatible compact smooth potential data. Problems 2–3 explicitly separate a charged background outside that class.

Exercise 10.1 (All residual shifts of Cauchy data). Let a smooth Lorenz-gauge Maxwell potential have initial data $(a_\phi, b_\phi, a_A, b_A)$ as in Lecture 10, with a fixed source ρ. Let χ be the smooth wave with arbitrary real compact smooth data $\chi(0) = p$, $\chi_t(0) = q$. Set $\phi'_{\mathrm{em}} = \phi_{\mathrm{em}} - \chi_t$, $\mathbf{A}' = \mathbf{A} + \nabla\chi$.

(a) Find the increments of all four potential data. Verify directly that both compatibility expressions $b_\phi + \operatorname{div} a_A$ and $-\Delta a_\phi - \operatorname{div} b_A - \rho(0)$ are unchanged. Compare the two initial electromagnetic fields.

(b) Prove that fixing all four potential data rules out every nonzero residual gauge wave in this compact-data class. Contrast this with fixing only the electromagnetic fields. If compactness of the gauge data is dropped, explain why a constant χ is physically trivial.

Hint. Use $\chi_{tt}(0) = \Delta p$ before evaluating the scalar-potential velocity. The vector displacement and velocity involve different gauge data.

Answer. The increments are $(-q, -\Delta p, \nabla p, \nabla q)$. Both compatibility expressions and both field data are unchanged. Unchanged potential data force $p = q = 0$ in the compact-data class, hence $\chi = 0$ by wave uniqueness; a global constant has $\mathrm{d}\chi = 0$.

Exercise 10.2 (A smooth charge with a Coulomb tail). Choose a radial nonnegative $b \in C_c^\infty(\mathbb{R}^3)$ with $\int b\,\mathrm{d}^3x = 1$ and support in $\overline{B_R(0)}$, $R > 0$. Fix real $Q \neq 0$ and prescribe the static source $\rho_Q(x) = Qb(x)$, $\mathbf{J} = 0$ for all time. For $r > 0$ define, writing $\rho_Q(s)$ for its radial profile,

$$M_Q(r) = 4\pi \int_0^r s^2 \rho_Q(s)\mathrm{d}s, \quad E_r(r) = \frac{M_Q(r)}{4\pi r^2}, \quad \phi_Q(r) = \int_r^\infty E_r(s)\mathrm{d}s.$$

(a) Prove that $E_Q(x) = E_r(|x|)x/|x|$ and $\phi_Q(|x|)$ extend smoothly across zero. Prove Gauss's law there as well as away from zero. Show that $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q, 0)$ solves Maxwell in Lorenz gauge.

(b) Determine the exact potential and field for $r \geq R$. Prove $\phi_Q \notin L^2$, hence $\phi_Q \notin H^1$, while $\nabla\phi_Q \in L^2$ and the field energy is finite. Calculate the electric energy outside B_R.

(c) Identify the failed hypotheses of Lecture 10's theorem. Why does conservation alone not force $Q = 0$ for this source?

Hint. Remove the radial denominator by proving $E_Q(x) = x\int_0^1 a^2\rho_Q(ax)\mathrm{d}a$. Its divergence is the integral of an a-derivative. For the potential, integrate this smooth field on the segment from zero to x.

Answer. The fields are smooth, with $\phi_Q(r) = Q/(4\pi r)$ and $E_r(r) = Q/(4\pi r^2)$ for $r \geq R$. The potential fails L^2 at infinity, but its gradient has finite norm. Exterior electric energy is $Q^2/(8\pi R)$. The source is spatially compact and conserved, but is not compact in time.

Exercise 10.3 (An affine charged evolution class). Keep the background of Exercise 10.2. Let $(\rho_1, J_1) \in C_c^\infty(\mathbb{R}^{1+3}; \mathbb{R} \times \mathbb{R}^3)$ be conserved. Give compact smooth perturbation data $(a_{1\phi}, b_{1\phi}, a_{1A}, b_{1A})$ satisfying

$$b_{1\phi} + \operatorname{div} a_{1A} = 0, \qquad -\Delta a_{1\phi} - \operatorname{div} b_{1A} = \rho_1(0).$$

Construct a smooth Lorenz-gauge solution for total source $(\rho_Q + \rho_1, J_1)$ and total initial data obtained by adding the static background. Prove its total charge is Q, its electric and magnetic fields have finite energy on each slice, and $\phi_{\mathrm{em}}(t) \in \phi_Q + H^1(\mathbb{R}^3)$ while $\phi_{\mathrm{em}}(t) \notin H^1$. State the uniqueness conclusion actually supplied by Lecture 10. Explain why this construction does not put the charged background in a Schwartz canonical phase space.

Hint. Apply the compact theorem only to the perturbation. Use its retarded construction and finite propagation to keep the perturbation data compact on each bounded time interval. Add the two Maxwell solutions afterwards.

Answer. Use Lecture 10 to construct (ϕ_1, A_1) and set $(\phi_{\mathrm{em}}, \mathbf{A}) = (\phi_Q + \phi_1, A_1)$. The perturbation, hence the total field after subtracting the fixed background, is unique in the theorem's C^2 Lorenz class with the specified potential data. The scalar potential belongs to the affine class $\phi_Q + H^1$, not to H^1.

Exercise 10.4 (Local potentials with no global real one-form). On the oriented unit sphere, with area form $\sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$, let $g_{\mathrm{mag}} \neq 0$ be real. On $U_N = S^2 \setminus \{\text{south pole}\}$ and $U_S = S^2 \setminus \{\text{north pole}\}$, respectively, take

$$A_N = g_{\mathrm{mag}}(1 - \cos\theta)\mathrm{d}\varphi, \qquad A_S = -g_{\mathrm{mag}}(1 + \cos\theta)\mathrm{d}\varphi.$$

These are spatial real one-form potentials; the task uses exterior calculus from *Topology and Geometry*, Part I, not bundle theory as a prerequisite.

(a) Verify smoothness at each included pole, compute the common curvature $F = \mathrm{d}A_N = \mathrm{d}A_S$, and find its total flux.

(b) Find a gauge function on a simply connected azimuthal sector in the overlap. Prove that there is no single-valued real gauge function for this shift on the whole overlap, and no globally smooth real one-form A on S^2 with $\mathrm{d}A = F$.

The interpretation by nontrivial bundles and single-valued charged transition phases, including the allowed flux quantization, is a later topic: *Topology and Geometry*, Lectures 20–22. No bundle construction or quantization theorem is assumed in this exercise.

Hint. Put $X = \sin\theta\cos\varphi$, $Y = \sin\theta\sin\varphi$ and $Z = \cos\theta$. Rewrite $\mathrm{d}\varphi$ using $X\mathrm{d}Y - Y\mathrm{d}X$. Test exactness on the equatorial circle and then apply Stokes on the closed sphere.

Answer. $F = g_{\mathrm{mag}} \sin\theta \mathrm{d}\theta \wedge \mathrm{d}\varphi$ has flux $4\pi g_{\mathrm{mag}} \neq 0$. Locally $A_N - A_S = \mathrm{d}(2g_{\mathrm{mag}}\varphi)$, but its equatorial integral is $4\pi g_{\mathrm{mag}}$, excluding a single-valued real gauge function on the overlap and a global real potential on the sphere.

Classical Field Theory — Exercise Sheet 11

Constraints and boundaries

Hints and answers

Use Lecture 11's canonical bracket, with $\mathbf{\Pi} = -\mathbf{E}$. A primary constraint is imposed only after forming the Hamilton equations. All field data below are real and smooth. The second problem uses Schwartz data on $\mathbb{R}^3$; the third explicitly changes the spatial region and its allowed boundary variations.

Exercise 11.1 (An entire constraint chain). Consider two mechanical coordinates $q(t)$ and $u(t)$ with Lagrangian

$$L(q, u, \dot{q}, \dot{u}) = \frac{1}{2}(\dot{q} - u)^2.$$

Use the following consistency procedure: find the image of the Legendre map, form the canonical Hamiltonian on that image, add an arbitrary multiplier for each primary constraint, and preserve the constraints until no new condition appears. The bracket on $(q, u; p_q, p_u)$ is the ordinary canonical bracket of mechanics.

(a) Carry out every step. State which conditions hold strongly and which hold only on the constraint surface. Classify the constraints.

(b) Find a time-dependent generator producing $\delta q = \epsilon(t)$ and $\delta u = \dot{\epsilon}(t)$ for an arbitrary smooth real function ϵ. Determine the transformation of the total-Hamiltonian multiplier needed to map solutions to solutions.

(c) At a fixed time, show that every two points on the constraint surface are gauge equivalent. How many canonical pairs remain on this quotient?

Hint. The relation for p_u contains no velocity. Preserve it before imposing it inside brackets. The values of ϵ and $\dot{\epsilon}$ can be prescribed independently at one time.

Answer. The constraints are $p_u = p_q = 0$, both first class. The total Hamiltonian is $p_q^2/2 + up_q + \lambda p_u$. The generator is $\epsilon p_q + \dot{\epsilon} p_u$ and $\lambda' = \lambda + \ddot{\epsilon}$. The quotient has no canonical pairs.

Exercise 11.2 (A mass term changes the constraint class). On $\mathbb{R}^{1+3}$ take the vacuum density

$$L_M = \frac{1}{2}|\mathbf{A}_t + \nabla\phi_{\mathrm{em}}|^2 - \frac{1}{2}|\operatorname{curl}\mathbf{A}|^2 + \frac{M^2}{2}(\phi_{\mathrm{em}}^2 - |\mathbf{A}|^2), \qquad M > 0.$$

At each time all canonical fields and the multiplier are Schwartz. Use the extended bracket of Lecture 11 and define $P[\eta] = \int \eta\pi_\phi \, \mathrm{d}^3x$ for real Schwartz η.

(a) Derive the momenta, canonical Hamiltonian and consistency chain. Does preserving the secondary constraint leave $\partial_t\phi_{\mathrm{em}}$ arbitrary?

(b) Put $\mathcal{D}_M = \operatorname{div}\mathbf{\Pi} - M^2\phi_{\mathrm{em}}$ and $C_M[\chi] = -\int \chi\mathcal{D}_M \, \mathrm{d}^3x$. Compute all primary/secondary brackets. For smearings with $\int \eta\chi = 1$, write the two-by-two off-diagonal bracket block and its inverse. Why does its inverse have no massless limit?

(c) Impose both constraints and express the scalar variables in terms of $\mathbf{A}$ and $\mathbf{\Pi}$. Pull back the canonical form and determine how many spatial component pairs remain. Contrast this with Maxwell's gauge quotient.

Hint. The mass term changes neither the velocity Hessian nor $\mathbf{\Pi}$. It does change the scalar-potential derivative of the Hamiltonian. Keep the boundary term before moving $\nabla\phi_{\mathrm{em}}$ onto $\mathbf{\Pi}$.

Answer. The primary constraint is $\pi_\phi = 0$, and the secondary one is $\mathcal{D}_M = 0$. Its preservation fixes $\partial_t\phi_{\mathrm{em}} = -\operatorname{div}\mathbf{A}$. The mixed bracket is $\{P[\eta], C_M[\chi]\} = -M^2\int\eta\chi$. Three spatial pairs remain; the longitudinal pair is not a Maxwell gauge direction.

Exercise 11.3 (A boundary generator need not vanish). Let $\Omega = B_1(0) \subset \mathbb{R}^3$. For this problem the phase variables $\mathbf{A}, \mathbf{\Pi}$ are smooth on a neighbourhood of $\overline{\Omega}$, with no fixed normal trace for $\mathbf{\Pi}$. Their canonical form is $\int_\Omega \delta A_i \wedge \delta\Pi_i \,\mathrm{d}^3x$. A functional has a *smooth bulk derivative* if its variation is an integral of smooth coefficients against arbitrary smooth field variations, without a boundary evaluation term. For a fixed smooth real χ, set

$$C_\Omega[\chi] = -\int_\Omega \chi \operatorname{div}\mathbf{\Pi}\,\mathrm{d}^3x, \qquad G_\Omega[\chi] = \int_\Omega \mathbf{\Pi}\cdot\nabla\chi\,\mathrm{d}^3x.$$

(a) Derive their relation, including the surface integral, and vary it. Explain why $C_\Omega[\chi]$ generally lacks a smooth bulk derivative when χ is nonzero on the boundary, while $G_\Omega[\chi]$ has one.

(b) Find the Hamiltonian vector of $G_\Omega[\chi]$. On the Gauss surface $\operatorname{div}\mathbf{\Pi} = 0$, must its value and its contraction with the constrained canonical form vanish?

(c) Evaluate both functionals and their boundary difference for $\mathbf{\Pi} = e_1$ and $\chi = x^1$. Identify the boundary assumption that distinguishes this problem from the all-space integrations by parts in Lecture 11.

Hint. Use the outward normal on the sphere. A compact interior test first determines a candidate bulk gradient. A free normal boundary variation then tests whether that candidate represents the whole derivative.

Answer. The differentiable completion is $G_\Omega = C_\Omega + \int_{\partial\Omega}\chi\mathbf{\Pi}\cdot n\,\mathrm{d}S$. Its vector is $(\nabla\chi, 0)$. In the example $C_\Omega = 0$ and $G_\Omega = 4\pi/3$, with the same nonzero boundary charge.

Classical Field Theory — Exercise Sheet 12

Completing transverse reduction

Hints and answers

Use the unitary Fourier transform of Lecture 4. The real canonical form is

$$\Omega((a,p),(b,q)) = \int (a\cdot q - b\cdot p).$$

The projector is $P_{\mathrm{T}}(k) = I - kk^{\mathsf{T}}/|k|^2$ for $k \neq 0$. All claims about a fixed annulus below concern the data and smearings specified in Lecture 12, not pointwise Fourier observables.

Exercise 12.1 (A finite Dirac bracket and an oblique mode). For a fixed $k \in \mathbb{R}^3 \setminus \{0\}$, consider the finite canonical model $(a,p) \in \mathbb{R}^3 \times \mathbb{R}^3$ with $\{a_i, p_j\} = \delta_{ij}$. Set $c_1 = k\cdot p$ and $c_2 = k\cdot a$. Given an invertible constraint matrix $C_{ab} = \{c_a, c_b\}$, its *Dirac bracket* is

$$\{F,G\}_D = \{F,G\} - \{F,c_a\}(C^{-1})_{ab}\{c_b,G\}.$$

This formula is supplied for this finite system.

(a) Compute C, its inverse, and every coordinate Dirac bracket. Check that each constraint has zero Dirac bracket with every coordinate.

(b) At $k = (1,2,2)$, display the projector and project $(3,-1,2)$. Recover the annulus smeared bracket of Lecture 12 from this matrix symbol, using real vector smearings f, g.

(c) At $k = 0$, which step fails? Explain why periodic real gauge parameters on $\mathbb{T}^3$ leave three constant canonical pairs, and give their form.

Hint. Compute the two off-diagonal constraint brackets before inverting. For smearings use symmetry and idempotence of the Fourier multiplier. A periodic gradient has zero mean.

Answer. The nonzero bracket is $\{a_i, p_j\}_D = P_{\mathrm{T}}(k)_{ij}$. At $(1,2,2)$ the matrix and projection are

$$\frac{1}{9}\begin{pmatrix} 8 & -2 & -2 \\ -2 & 5 & -4 \\ -2 & -4 & 5 \end{pmatrix}, \qquad (22,-19,8)/9.$$

The zero-mode form is $(2\pi)^3 \sum_i \delta a_i \wedge \delta p_i$.

Exercise 12.2 (A union of annuli, rather than one annulus). For $n \geq 2$ let $\mathcal{U}_n = \{n^{-1} < |k| < n\}$ and let $\mathcal{Y}$ be the union of the real Schwartz classes with smooth compact Fourier support in $\mathcal{U}_n$.

(a) Prove that $\mathcal{Y}$ is dense in the real spaces $L^2(\mathbb{R}^3;\mathbb{R})$ and $H^1(\mathbb{R}^3;\mathbb{R})$, using the appropriate norm in each case. You may use the usual L^2 density of compact smooth functions, proved by truncation and smoothing.

(b) Prove the corresponding density for transverse vector fields, with approximants $P_{\mathrm{T}}v_n$. Keep the real-field Fourier condition.

(c) Show that the analogous class in one fixed annulus is not dense in either whole space. Explain why a bounded extension from a dense union must not be justified by density of just one annulus.

Hint. Cut off low and high frequencies, smooth inside the resulting annulus, and average a profile with its conjugate reflection. Fourier weights are bounded above and below on each compact annulus.

Answer. The union is dense in both norms, also after transverse projection. One fixed annulus misses every nonzero profile supported in a disjoint frequency shell and therefore is not dense.

Exercise 12.3 (A gauge parameter outside L^2). Choose a real radial $\eta \in C_c^\infty(\mathbb{R}^3)$ equal to one on $|k| \le 1$. Define a tempered scalar distribution and a vector by their Fourier profiles

$$\widehat{\chi}_0(k) = \frac{\eta(k)}{|k|^2}, \qquad \widehat{A}(k) = ik\,\widehat{\chi}_0(k) \quad (k \neq 0).$$

A locally integrable function of at most polynomial growth defines a tempered distribution. You may also use that a distribution with all first derivatives zero is constant.

(a) Show that χ_0 is well-defined and real, that $A \in H^1$, and that $\chi_0 \notin L^2$. Can any L^2 scalar have gradient A?

(b) For a general real $A \in H^1(\mathbb{R}^3;\mathbb{R}^3)$ compare the two profiles $\widehat{\chi}_L = -i(k\cdot\widehat{A})/|k|^2$ and $-\widehat{\chi}_L$. Which one has gradient $(I - P_{\mathrm{T}})A$, and which gauges A to $P_{\mathrm{T}}A$? Justify that χ_L is tempered even near zero frequency.

Hint. Use the radial measure $r^2 \mathrm{d}r\mathrm{d}\Omega$ when testing integrability. For the general parameter use Cauchy–Schwarz with $|k|^{-1}$ near zero, and test against Schwartz functions at infinity.

Answer. The gradient A belongs to H^1 although its parameter has no L^2 representative, even modulo constants. The parameter χ_L represents the longitudinal part; the gauge transformation uses $-\chi_L$.

Exercise 12.4 (Weak symplectic does not mean every gradient exists). On the real space $\mathcal{E}_T = H^1_{\mathrm{trans}} \times L^2_{\mathrm{trans}}$, use the canonical form from Lecture 12. Let $\theta(r)$ be smooth, zero for $r \le 1$ and one for $r \ge 2$, and define the real transverse L^2 vector h by

$$\widehat{h}(k) = \theta(|k|)|k|^{-2}P_{\mathrm{T}}(k)e_1.$$

(a) Prove that $h \in L^2$ but $h \notin H^1$. Show that $F(a,p) = \int h \cdot p\,\mathrm{d}^3x$ is continuous on $\mathcal{E}_T$.

(b) Show that no $X \in \mathcal{E}_T$ satisfies $\iota_X\Omega = DF$. Why does this not contradict the nondegeneracy proved in Lecture 12?

(c) Does finiteness of $\int(|\operatorname{curl} A|^2 + |\Pi|^2)$ imply membership in this inhomogeneous phase space? Test a nonzero constant potential with zero momentum.

Hint. The angular integral of $|P_{\mathrm{T}}(n)e_1|^2$ is $8\pi/3$. Test the proposed Hamiltonian vector against arbitrary transverse momentum variations before considering its other component.

Answer. The high-frequency radial integrals are proportional to $\int_2^\infty r^{-2}\mathrm{d}r$ for L^2 and $\int_2^\infty (1+r^2)r^{-2}\mathrm{d}r$ for H^1. A representing vector would require $X_A = h$, impossible in H^1. Finite curl energy alone does not imply $A \in L^2$.

Classical Field Theory — Exercise Sheet 13

Radiation estimates and boundary conditions

Hints and answers

Use Lecture 13's real Maxwell fields, retarded power theorem and amplitude $\mathcal{A}(u,n) = (4\pi)^{-1}\int \mathbf{J}(u+n\cdot y, y)\mathrm{d}^3y$. Here $n\in\mathbb{S}^2$, $u=t-r$, and $P_{\mathrm{T}}(n)=I-nn^{\mathsf{T}}$. The third problem also uses Lecture 6's time-reflected advanced kernel. The singular-source formula in the last problem is supplied separately.

Exercise 13.1 (A general slow-time source). Let $\rho_0(s,y)$ be scalar and $J_0(s,y)$ vector-valued, both real and smooth with compact spacetime support, satisfying $\partial_s\rho_0+\operatorname{div}_y J_0=0$. For $0<\varepsilon\le 1$ put

$$\rho_\varepsilon(t,y)=\rho_0(\varepsilon t,y),\qquad \mathbf{J}_\varepsilon(t,y)=\varepsilon J_0(\varepsilon t,y),\qquad p_0(s)=\int y\rho_0(s,y)\mathrm{d}^3y.$$

Define finite constants

$$M_0=\int \sup_s|\partial_s J_0(s,y)|\mathrm{d}^3y,$$
$$M_1=\int |y|\sup_s|\partial_s^2 J_0(s,y)|\mathrm{d}^3y,$$
$$M_2=\frac12\int |y|^2\sup_s|\partial_s^3 J_0(s,y)|\mathrm{d}^3y.$$

(a) Verify conservation and prove $p_0'(s)=\int J_0(s,y)\mathrm{d}^3y$. Find the scaled dipole $p_\varepsilon(t)=\int y\rho_\varepsilon(t,y)\mathrm{d}^3y$ and its second derivative.

(b) Prove, uniformly for all $u\in\mathbb{R}$, the explicit absolute estimate

$$\left|P_\varepsilon(u)-\frac{|\ddot p_\varepsilon(u)|^2}{6\pi}\right|\le\frac{2M_0M_1+M_1^2}{4\pi}\varepsilon^5.$$

(c) Improve the integrated error to $O(\varepsilon^6)$ by retaining the first delayed-sampling correction. Give an explicit bound using M_0,M_1,M_2. Why does this improvement not assert that the directional power density has an $O(\varepsilon^6)$ error?

Hint. Taylor-expand $\partial_s J_0(\varepsilon u+\varepsilon n\cdot y,y)$ at εu before squaring. The first correction is odd in n, whereas the transverse projector is even.

Answer. The dipole is $p_\varepsilon(t)=p_0(\varepsilon t)$. An $O(\varepsilon^6)$ bound is $\varepsilon^6(2M_0M_2+M_1^2+2M_1M_2+M_2^2)/(4\pi)$. The $O(\varepsilon^5)$ cross term cancels only after angular integration.

Exercise 13.2 (A nonradial source at zero acceleration). Choose a nonnegative $f\in C_c^\infty(B_1(0))$ with $\int f=1$ and nonzero first moment $\mu=\int yf(y)\mathrm{d}^3y$. Let $a\in\mathbb{R}^3\setminus\{0\}$ and choose a smooth compactly supported p_0 equal to as^3 on $|s|\le 1$. Use the conserved dipole source

$$p(t)=p_0(\varepsilon t),\qquad \rho=-p(t)\cdot\nabla f,\qquad \mathbf{J}=\dot p(t)f,\qquad 0<\varepsilon\le 1.$$

Compute the exact retarded power $P(0)$. Show that it is positive although $\ddot p(0)=0$, and evaluate it for $a=e_3$, $\mu=e_3/4$. Explain what this rules out about a relative error estimate. You may use the angular moments

$$\int n_in_j\mathrm{d}\Omega=\frac{4\pi}{3}\delta_{ij},\qquad \int n_in_jn_kn_l\mathrm{d}\Omega=\frac{4\pi}{15}(\delta_{ij}\delta_{kl}+\delta_{ik}\delta_{jl}+\delta_{il}\delta_{jk}).$$

Hint. Every sampled slow time $\varepsilon n \cdot y$ lies in the polynomial region. Keep the first moment of f rather than replacing it by a radial average.

Answer.

$$P(0) = \frac{6\varepsilon^6}{5\pi}\left(2|a|^2|\mu|^2 - (a\cdot\mu)^2\right) > 0.$$

For the specified aligned vectors it is $3\varepsilon^6/(40\pi)$.

Exercise 13.3 (The half-retarded, half-advanced field). For a real smooth compact spacetime conserved source, let $(\phi_{\mathrm{em}}^{\mathrm{ret}}, \mathbf{A}_{\mathrm{ret}})$ be its retarded potentials. Define the advanced potentials by replacing $t - |x-y|$ by $t + |x-y|$ in the same $1/(4\pi|x-y|)$ integrals. Time reflection of Lecture 6's kernel gives these smooth sourced solutions. Let the symmetric potentials be half the sum of the two, and form their Maxwell fields.

(a) At future null infinity, $t = r + u$, show that the advanced field is exactly zero for all sufficiently large r (at each fixed u). Find the symmetric field's limiting outward power in terms of the retarded power.

(b) At past null infinity, $t = v - r$, define

$$\overline{\mathcal{A}}(v,n) = \frac{1}{4\pi}\int \mathbf{J}(v - n\cdot y, y)\mathrm{d}^3y, \qquad \overline{\mathcal{V}}(v,n) = \frac{1}{4\pi}\int \rho(v - n\cdot y, y)\mathrm{d}^3y.$$

Derive the leading advanced electric and magnetic coefficients and the signed outward power of the symmetric field. Primes now mean ∂_v. Does averaging the potentials cancel radiation at future null infinity?

Hint. For the advanced field the source time is $t + |rn - y|$. At fixed t, its leading radial derivative has the opposite sign from the retarded case. Use conservation to relate the two advanced amplitudes.

Answer. Future outward power is $P_{\mathrm{ret}}(u)/4$. The advanced coefficients at past infinity are $E_* = -P_{\mathrm{T}}\overline{\mathcal{A}}'$ and $B_* = +n\times\overline{\mathcal{A}}' = -n\times E_*$. The symmetric past outward power is $-\frac{1}{4}\int |n\times\overline{\mathcal{A}}'|^2\mathrm{d}\Omega$.

Exercise 13.4 (Liénard–Wiechert: a supplied singular-source formula). Let $z\colon \mathbb{R}\to\mathbb{R}^3$ be a smooth complete worldline with $|\dot z(s)| \le v_* < 1$. Fix an event (t,x) with $x \neq z(t)$. Let $Q\in\mathbb{R}$ be its charge. For this problem only, take as given the point-source potential prescription

$$\phi_{\mathrm{em}}(t,x) = \frac{Q}{4\pi}\int \frac{\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s, \qquad \mathbf{A}(t,x) = \frac{Q}{4\pi}\int \frac{\dot z(s)\delta(t-s-|x-z(s)|)}{|x-z(s)|}\mathrm{d}s.$$

The delta integrals are understood locally at their roots. You may use the one-variable identity $\delta(g(s)) = \sum_j \delta(s-s_j)/|g'(s_j)|$ for simple roots.

(a) Prove existence and uniqueness of the retarded root $t = s_{\mathrm{ret}} + |x - z(s_{\mathrm{ret}})|$. Evaluate the two potentials in terms of $R = |x - z(s_{\mathrm{ret}})|$, $n = (x - z(s_{\mathrm{ret}}))/R$ and $v = \dot z(s_{\mathrm{ret}})$.

(b) For $z(s) = (3s/5, 0, 0)$ at $(t,x) = (1,0)$, compute the retarded time, Jacobian and scalar potential. Explain exactly which smooth-source hypotheses fail and why this calculation establishes no self-force law.

Hint. The distance function is v_*-Lipschitz in s. This proves global monotonicity even if $x = z(s)$ at a different time. Only the actual retarded root must have nonzero distance.

Answer.

$$\phi_{\mathrm{em}} = \frac{Q}{4\pi R(1 - n\cdot v)}, \qquad \mathbf{A} = \phi_{\mathrm{em}} v.$$

The example gives $s_{\mathrm{ret}} = 5/8$, $1 - n\cdot v = 8/5$, and $\phi_{\mathrm{em}} = 5Q/(12\pi)$.

Classical Field Theory — Exercise Sheet 14

First-order systems and real structures

Hints and answers

All spinor entries commute. Use Lecture 14's Hermitian inner product, unitary Fourier convention and real form $\Omega(h,g) = 2\,\mathrm{Im}\int h^\dagger g$. Problems 1 and 3 supply new four-component matrices; Problem 4 returns to the actual two-component matrices of Lecture 14.

Exercise 14.1 (Four components in three space dimensions). Let I_2 be the two-by-two identity and put

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -\mathrm{i} \\ \mathrm{i} & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \qquad \Gamma^0 = \begin{pmatrix} I_2 & 0 \\ 0 & -I_2 \end{pmatrix}, \quad \Gamma^i = \begin{pmatrix} 0 & \sigma_i \\ -\sigma_i & 0 \end{pmatrix}.$$

Define $\beta_D = \Gamma^0$, $\alpha_D^i = \Gamma^0\Gamma^i$ and $H_D(k) = \alpha_D^i k_i + m\beta_D$, where $m \geq 0$.

(a) Verify every Clifford anticommutator for signature $(+,-,-,-)$. Prove that $H_D(k)$ is Hermitian and find its eigenvalues, multiplicities and orthogonal spectral projectors, including the exceptional $m = k = 0$ case.

(b) For $\mathrm{i}\psi_t = (-\mathrm{i}\alpha_D^i\partial_i + m\beta_D)\psi$, derive the conserved current $\rho = \psi^\dagger\psi$, $j^i = \psi^\dagger\alpha_D^i\psi$ and prove $|n\cdot j| \leq \rho$ for every unit vector n. Explain why verifying these matrices alone does not prove the spin representation or Lorentz-covariance theorem cited in Lecture 14.

Hint. First multiply the Pauli matrices. Square H_D before finding its projectors, and use the trace to determine their ranks.

Answer. For $\nu = \sqrt{|k|^2 + m^2} > 0$, the eigenvalues are $\pm\nu$, each twice, with projectors $(I_4 \pm H_D/\nu)/2$. At $m = k = 0$, $H_D = 0$. The current obeys $\partial_t\rho + \partial_i j^i = 0$ and $|n\cdot j| \leq \rho$.

Exercise 14.2 (A constant symmetric-hyperbolic system). Let $A^1, \dots, A^d$ be constant Hermitian N-by-N matrices and B a constant skew-Hermitian matrix. For complex columns consider

$$w_t + A^i\partial_i w + Bw = 0, \qquad c_* = \sup_{|n|=1} \|n_i A^i\|_{\mathrm{op}}.$$

Here the operator norm is the Euclidean matrix norm.

(a) Construct the unique distributional solution in $C(\mathbb{R}; L^2)$ for each L^2 datum and prove norm preservation. Derive the smooth local current.

(b) Using Lecture 5's moving-ball argument, prove that data supported in a closed set K remain supported in $\{x : \mathrm{dist}(x,K) \leq c_*|t|\}$. Give the limiting argument for rough data and include $c_* = 0$.

(c) For $d = N = 2$, take $A^1 = \sigma_1$, $A^2 = 2\sigma_3$ and $B = \mathrm{i}\sigma_2$. Compute c_* and exhibit a column and normal attaining the current bound. Would Hermitian $B = I_2$ preserve the same norm?

Hint. The Fourier generator $\mathrm{i}k_iA^i + B$ is skew-Hermitian. On a shrinking ball use $j\cdot n \geq -c_*\rho$ at the boundary. For rough data pass local squared norms, rather than boundary traces, to the limit.

Answer.

$$\widehat{w}(t,k) = e^{-t(\mathrm{i}k_i A^i + B)}\widehat{w}(0,k), \qquad \rho = w^\dagger w, \quad j^i = w^\dagger A^i w.$$

The example has $c_* = 2$, attained by $w = e_1$, $n = e_2$. For $B = I_2$ the total squared norm decays as e^{-2t}.

Exercise 14.3 (Weyl sectors and a mass coupling). Use the four-component matrices of Problem 1. Define $\Gamma^5 = \mathrm{i}\Gamma^0\Gamma^1\Gamma^2\Gamma^3$ and $P_{\mathrm{ch},\pm} = (I_4 \pm \Gamma^5)/2$. Compute Γ^5 and its two eigenspaces. Write the Dirac Hamiltonian in the coordinates

$$\psi = \frac{1}{\sqrt{2}}\begin{pmatrix} u+v \\ u-v \end{pmatrix}, \qquad u, v \in \mathbb{C}^2.$$

Determine when the two chiral subspaces are invariant under evolution, and derive the two Weyl equations in the massless case.

Hint. Compare $[\Gamma^5, \alpha_D^i]$ with $\Gamma^5\beta_D + \beta_D\Gamma^5$. The displayed change of coordinates is unitary.

Answer.

$$\Gamma^5 = \begin{pmatrix} 0 & I_2 \\ I_2 & 0 \end{pmatrix}, \qquad H_{\mathrm{ch}}(k) = \begin{pmatrix} \sigma\cdot k & mI_2 \\ mI_2 & -\sigma\cdot k \end{pmatrix}.$$

The separate chiral subspaces are invariant exactly when $m = 0$.

Exercise 14.4 (A commuting Majorana-type real subspace). Return to Lecture 14's two-component matrices $\alpha = -\sigma_3$, $\beta = \sigma_1$ and define the real-linear, antilinear-in-complex-scalars map

$$C\psi = \sigma_3\overline{\psi}.$$

Prove $C^2 = I$, characterize its fixed space in L^2, and show that Dirac evolution preserves it. Derive the evolution of its two real entries. Restrict the real form Ω to that fixed space. Is the result nondegenerate? Explain why this calculation does not give a classical commuting replacement for a quantum Majorana field or its statistics.

Hint. The matrix σ_3 commutes with α and anticommutes with β. Test the imaginary part of the inner product of two fixed columns.

Answer. Fixed columns have the form $\psi = (a, \mathrm{i}b)^T$ with a, b real, and obey $a_t = a_x + mb$, $b_t = -b_x - ma$. The restricted form is identically zero.

Classical Field Theory — Exercise Sheet 15

Beyond the one-dimensional cubic proof

Hints and answers

Use the mass-one Klein–Gordon group $U_d(t)$ on $\mathcal{E}_d = H^1(\mathbb{R}^d) \times L^2(\mathbb{R}^d)$, with Hilbert norm $\|(f,g)\|_{\mathrm{lin}}^2 = \|f\|_{H^1}^2 + \|g\|_2^2$. Lecture 15 proves the forced estimate in one dimension; Problem 2 checks its dimension-independent Fourier proof. All fields below are real.

Exercise 15.1 (A general potential and an affine background). Let $V \in C^3(\mathbb{R})$ and let b be a smooth bounded solution of $-b'' + V'(b) = 0$ on $\mathbb{R}$. Do not assume $b \in L^2$. For $\phi = b + v$ consider $\phi_{tt} - \phi_{xx} + V'(\phi) = 0$.

(a) Derive the mass-one perturbation force $N_b(v)$, and show that it maps $H^1(\mathbb{R})$ locally Lipschitzly to $L^2(\mathbb{R})$, vanishing at zero. Give a constant on a ball of radius R using the maximum of $|V''|$ on a specified compact interval.

(b) For every initial ball $\|(v_0, g_0)\|_{\mathrm{lin}} \le B$, $B \ge 1$, give a common contraction radius and positive time. Prove existence, uniqueness among weak energy solutions, continuous dependence, and the finite-lifespan alternative. Which derivatives of V did the estimates use?

Hint. Subtract the stationary equation before asking for an L^2 force. On the bounded field range use the integral of V'' along the segment between two field values, then use Lecture 15's forcing lemma.

Answer.

$$N_b(v) = v - [V'(b+v) - V'(b)], \qquad L_R = 1 + \max_{|s| \le \|b\|_\infty + R/\sqrt{2}} |V''(s)|.$$

Take radius $2B$ and time $1/(2L_{2B})$. The estimate needs only $V \in C^2$.

Exercise 15.2 (The cubic survives in three dimensions). Take as given the Sobolev estimate $\|u\|_{L^6(\mathbb{R}^3)} \le C_S \|u\|_{H^1(\mathbb{R}^3)}$ for every real $u \in H^1$, where C_S is a fixed positive constant. Consider

$$u_{tt} - \Delta u + u^3 = 0, \qquad (u(0), u_t(0)) \in \mathcal{E}_3.$$

(a) Justify the three-dimensional version of the forced mass-one energy estimate, including the weak equation. Prove local existence, energy-class uniqueness and continuous dependence by a Duhamel contraction, with a common lifespan for a bounded initial ball.

(b) Would the same $H^1 \to L^2$ force argument handle u^5? For a nonzero real $\eta \in C_c^\infty(\mathbb{R}^3)$, compute the scaling of $u_\Lambda(x) = \Lambda^{1/2}\eta(\Lambda x)$, $\Lambda \ge 1$, in H^1, and of its cubic and quintic forces in L^2. Limit the conclusion to this contraction argument.

Hint. In Fourier space the linear rotation uses $\nu(k) = \sqrt{1 + |k|^2}$ in every dimension. For the cubic difference apply Hölder with three L^6 factors rather than an L^∞ factor.

Answer. The reference force is $N(u) = u - u^3$, with $L_R = 1 + 3C_S^3 R^2$. For $B \ge 1$, take $R = 2B$, $T = 1/(2L_R)$. The quintic L^2 norm scales as Λ, while the cubic norm is fixed and the H^1 norms stay bounded.

Exercise 15.3 (Sine-Gordon on a weighted path space). Let b be a smooth bounded stationary sine-Gordon background, $-b'' + \sin b = 0$, and write $\phi = b + v$. For any fixed $T > 0$, equip $X_T = C([0,T]; \mathcal{E}_1)$ with the equivalent norm

$$\|z\|_a = \sup_{0 \le t \le T} e^{-at} \|z(t)\|_{\mathrm{lin}}, \qquad a > 0.$$

Derive the reference mass-one Duhamel force. Show that the Duhamel map is a contraction on the entire space X_T for $a = 4$, with no radius restriction. Prove existence and uniqueness on every finite interval and give a continuous-dependence bound. Why is an amplitude-dependent local constant unnecessary? Do not assume that b or $\sin b$ belongs to L^2.

Hint. The derivative of $v - \sin(b+v) + \sin b$ lies between zero and two. The weighted norm estimates a time integral by integrating $e^{-a(t-s)}$.

Answer. $N_b(v) = v - \sin(b+v) + \sin b$ has global Lipschitz constant two into L^2. The weighted contraction factor is at most $2/a = 1/2$. For fixed data differences the unweighted solution difference is at most $2e^{4T}$ times their energy norm.

Exercise 15.4 (Defocusing continuation with the analytic input stated). For the three-dimensional cubic solution of Problem 2, assume additionally that the following energy identity holds throughout its maximal energy-class lifespan:

$$E_0 = \int_{\mathbb{R}^3} \left[\frac{1}{2} u_t^2 + \frac{1}{2} |\nabla u|^2 + \frac{1}{4} u^4 \right] \mathrm{d}^3 x.$$

This is an additional hypothesis at the weak energy regularity, not a consequence of L3's classical balance. Prove that the initial integral is finite and that the solution is global. Give a bound for its full $\mathcal{E}_3$ norm on every finite time interval, explicitly recovering the L^2 displacement term.

Hint. Interpolate L^4 between L^2 and the supplied L^6 bound. The missing L^2 estimate follows from the norm primitive of u_t.

Answer.

$$\|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \le \left(\|u(0)\|_2 + |t| \sqrt{2E_0} \right)^2 + 4E_0.$$

The continuation alternative therefore excludes every finite endpoint.

Classical Field Theory — Exercise Sheet 16

Energy, plateaux and local blow-up

Hints and answers

Use the general one-dimensional local theorem proved on Sheet 15: for $V \in C^3$, $V'(0) = 0$, data in $H^1(\mathbb{R}) \times L^2(\mathbb{R})$ have a unique weak energy solution, and a finite maximal endpoint forces its full energy-space norm to tend to infinity. Its reference linear mass is one. The conservation argument below must be made at that weak regularity.

Exercise 16.1 (A nonnegative potential beyond the quartic). Let $V \in C^3(\mathbb{R})$ satisfy $V \geq 0$ and $V(0) = V'(0) = 0$. For the energy solution of $u_{tt} - u_{xx} + V'(u) = 0$ with data (u_0, g_0), set

$$E(t) = \int_{\mathbb{R}} \left[\frac{1}{2} u_t^2 + \frac{1}{2} u_x^2 + V(u) \right] \mathrm{d}x.$$

(a) Prove finiteness, continuity and conservation of E by smoothing the actual solution in space. Justify the potential chain rule and the endpoint limits. Do not assume a density theorem for nonlinear solutions.

(b) Prove global existence and give a finite-time bound on the full $H^1 \times L^2$ norm. For $V(s) = s^2/2 + s^6/6$, obtain a stronger bound uniform in time. Identify which part of the proof needs $V \geq 0$.

Hint. On a compact lifespan interval, $H^1(\mathbb{R})$ bounds the range of u. Use $|V(a) - V(b)| \leq C_M(|a| + |b|)|a - b|$ there. For a fixed smoothing parameter, Taylor's remainder is quadratic in the L^2 time increment; the smoothed field has enough norm derivatives.

Answer. Energy is constant and

$$\|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \leq \left(\|u_0\|_2 + |t|\sqrt{2E_0} \right)^2 + 4E_0.$$

For the stated sextic potential the stronger bound is $\|(u(t), u_t(t))\|_{\mathrm{lin}}^2 \leq 2E_0$.

Exercise 16.2 (Sine-Gordon energy does not bound every phase norm). Take $V(u) = 1 - \cos u$ and background zero.

(a) Use Problem 1 to prove energy conservation and global energy-class evolution. Verify explicitly the initial potential integrability and continuity needed for that argument.

(b) Choose a fixed nondecreasing $\eta \in C^\infty(\mathbb{R})$, zero for $s \leq 0$ and one for $s \geq 1$. For $R \geq 1$ take initial displacement

$$u_R(x) = 2\pi\eta(x + R + 1)\eta(R + 1 - x)$$

and zero velocity. Compute its energy in terms of two integrals on $[0, 1]$ and show that it is independent of R, while $\|u_R\|_2 \to \infty$. Does this contradict global existence or the finite-time bound of Problem 1?

Hint. Use $0 \leq 1 - \cos s \leq s^2/2$ and $|\sin s| \leq |s|$. The plateau has zero potential, but its two fixed transition ramps do not.

Answer.

$$E[u_R, 0] = 4\pi^2 \int_0^1 \eta'(s)^2 \mathrm{d}s + 2 \int_0^1 [1 - \cos(2\pi\eta(s))]\mathrm{d}s, \qquad \|u_R\|_2^2 \geq 8\pi^2 R.$$

Each datum evolves globally, but this fixed energy does not uniformly control the displacement norm of the entire family.

Exercise 16.3 (Focusing blow-up from compact data on the line). Change the potential to $V(u) = -u^4/4$ on $\mathbb{R}$, so $u_{tt} - u_{xx} = u^3$. Fix $0 < T < R$ and choose a real $\chi \in C_c^\infty(\mathbb{R})$ with $0 \leq \chi \leq 1$ and $\chi = 1$ on $[-R, R]$. Take

$$u_0 = \frac{\sqrt{2}}{T}\chi, \qquad g_0 = \frac{\sqrt{2}}{T^2}\chi.$$

For this conditional exercise, assume additionally that the unique energy solution for the displayed data has a C^2 representative throughout its maximal energy-class lifespan $[0, T_*)$. This is a regularity hypothesis, and assumes no global existence.

(a) Compute its finite initial focusing energy. Compare on the cone $|x| \leq R - t$ with $q(t) = \sqrt{2}/(T - t)$ for every common time $0 \leq t < \min\{T, T_*\}$. Prove the needed local uniqueness rather than assuming a global finite-energy theorem for q.

(b) Deduce $T_* \leq T$. State carefully what was proved if the maximal solution terminates earlier than T. Why do these data improve on simply placing the spatially constant ODE on all of $\mathbb{R}$?

Hint. For $w = u - q$, factor $u^3 - q^3$ and use a positive mass-one energy on the shrinking interval. Its derivative is at most a bounded coefficient times that energy; zero initial energy then stays zero.

Answer.

$$E_0 = T^{-2} \int \chi'^2 \mathrm{d}x + T^{-4} \int (\chi^2 - \chi^4)\mathrm{d}x > 0.$$

On the common cone $u = q$, so a solution extending past T is impossible. The maximal energy lifespan is finite, with $T_* \leq T$.

Classical Field Theory — Exercise Sheet 17

Forced and accidental massless directions

Hints and answers

Use the real scalar kinetic coefficient $1/2$ and the orbit–Hessian lemma of Lecture 17. Counts of massless modes below refer to the linearised field. A vacuum background need not belong to L^2; perturbation data may be Schwartz or in the earlier energy class.

Exercise 17.1 (The full orthogonal orbit). For $N \geq 3$, $\lambda, v > 0$, consider

$$V(u) = \frac{\lambda}{8}(|u|^2 - v^2)^2, \qquad u \in \mathbb{R}^N, \qquad u_0 = vn, \quad |n| = 1,$$

with the usual $O(N)$ action and canonical kinetic term. Find the vacuum set, the stabiliser at u_0, its dimension, and the orbit tangent. Derive the Hessian and diagonalise the linearised field using the radial and tangential projectors, without first rotating n to a coordinate axis. Identify which masses are symmetry-forced. Finally discuss $N = 1$ separately: does its broken discrete symmetry force a massless mode?

Hint. An orthogonal transformation fixing n restricts to an arbitrary orthogonal transformation on $n^\perp$. Compute the gradient before substituting the vacuum.

Answer. The stabiliser is $O(N-1)$, the orbit is S^{N-1}_v, and $D^2V(u_0) = \lambda v^2 nn^{\mathsf{T}}$. There is one mass $\sqrt{\lambda}v$ and $N-1$ forced massless components. At $N = 1$ there are no tangent modes and the mass remains $\sqrt{\lambda}v$.

Exercise 17.2 (A flat singlet still has a nonlinear restoring force). Add a real scalar z with its own canonical kinetic term and potential

$$\widetilde{V}(u, z) = \frac{\lambda}{8}(|u|^2 - v^2)^2 + z^4.$$

Let $O(N)$ act on u and trivially on z. Find the vacuum set, the Hessian kernel at $(vn, 0)$, and the subspace forced by the symmetry. Can the extra singlet direction be tangent to a curve of vacua? On the spatial torus, set $u = vn$ and take a homogeneous singlet $z(0) = A > 0$, $z_t(0) = 0$. Derive its nonlinear period as a definite integral and its dependence on A. Explain how a zero linearised mass is compatible with this nonlinear oscillation. These homogeneous data are not claimed to be L^2 data on $\mathbb{R}^d$.

Hint. The nonnegative summands must both vanish at a minimum. Use the conserved mechanical energy $z_t^2/2 + z^4$ and integrate one quarter of the periodic orbit between zero and a turning point.

Answer. The vacuum set is $S^{N-1}_v \times \{0\}$; the Hessian kernel has dimension N, but only $N-1$ directions are symmetry-forced. The singlet period is

$$T_A = \frac{2\sqrt{2}}{A} \int_0^1 \frac{\mathrm{d}s}{\sqrt{1 - s^4}}.$$

Exercise 17.3 (An ineffective group factor belongs to the stabiliser). Take $G = SO(3)\times U(1)$ acting on a real triplet by $(R, e^{i\theta})u = Ru$. Use the radial potential of Problem 1 with $N = 3$ and vacuum $u_0 = v(2,-1,2)/3$. Determine the full stabiliser and the differential of the orbit map $(c,\theta) \mapsto c \times u_0$ on $\mathbb{R}^3 \oplus \mathbb{R}$. Find a basis of its kernel, the number of broken directions and the Hessian nullity. Why would counting $\dim G$ or only counting rotations that fix u_0 give a wrong answer?

Hint. The entire $U(1)$ factor acts trivially. A rotation generator fixes a nonzero vector exactly when its axial vector is parallel to that vector.

Answer. $H \cong SO(2) \times U(1)$ has dimension two, while $\dim G = 4$. The kernel is spanned by $((2,-1,2),0)$ and $(0,1)$, where the latter has zero rotation part. There are two broken and two massless directions.

Exercise 17.4 (Kinetic normalisation before diagonalising masses). For a real three-component perturbation h, consider the quadratic density

$$L = \frac{1}{2}\partial_\mu h^{\mathsf{T}} K \partial^\mu h - \frac{1}{2} h^{\mathsf{T}} M h, \qquad K = \begin{pmatrix} 2 & 1 & 0 \\ 1 & 2 & 1 \\ 0 & 1 & 1 \end{pmatrix}, \quad M = \begin{pmatrix} 4 & 0 & 0 \\ 0 & 9 & 9 \\ 0 & 9 & 9 \end{pmatrix}.$$

The real phase data may be taken Schwartz. Verify the factorisation $K = S^{\mathsf{T}} S$ for

$$S = \begin{pmatrix} 1 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix}.$$

Derive the canonical momentum, and find the corresponding change of coordinates and momenta that preserves $\int \delta h_a \wedge \delta p_a$. Compute the physical squared masses and compare them with the raw eigenvalues of M. State the general eigenproblem and explain what happens to the kernel dimension under this kinetic normalisation.

Hint. Use $Q = Sh$ and solve for the new momentum from $p^{\mathsf{T}}\delta h$. Transform the potential by congruence, not by similarity of M alone.

Answer.

$$p = K h_t, \qquad Q = Sh, \qquad P = S^{-\mathsf{T}} p, \qquad S^{-\mathsf{T}} M S^{-1} = \operatorname{diag}(0,4,9).$$

The squared masses are $0, 4, 9$; the raw Hessian eigenvalues are $0, 4, 18$.

Classical Field Theory — Exercise Sheet 18

Spectrum and moving kinks

Hints and answers

Retain $b(x) = \tanh x$, $V(s) = \frac{1}{2}(1-s^2)^2$ and $H_K = -\partial_x^2 + 4 - 6\operatorname{sech}^2 x$ with domain $H^2 \subset L^2$. Lecture 18 proves $H_K = A^\dagger A$ for $A = \partial_x + 2\tanh x$ and $\ker H_K = \operatorname{span}\{\operatorname{sech}^2 x\}$. Use its H^1 form $\mathfrak{q}[h] = \|Ah\|_2^2$ and density/integration-by-parts results.

Exercise 18.1 (The full spectrum through two adjoint pairs). The following analytic spectral theorem is given for this problem: if W is real, bounded and smooth, with $W(x) \to 0$ at both ends, then $-\partial_x^2 + 4 + W$ on $H^2(\mathbb{R})$ is self-adjoint, its essential spectrum is $[4, \infty)$, and every spectral point outside that interval is an isolated finite-multiplicity eigenvalue. This follows from Teschl, *Mathematical Methods in Quantum Mechanics* (2009), Lemmas 9.33 and 9.35, pp. 215–216, after shifting by four; the discrete/essential spectrum convention is in §6.4, p. 145. This input supplies more than the formal factorisation in Lecture 18. Set $B = \partial_x + \tanh x$ with domain H^1.

(a) Find $B^\dagger$ and the product domains, and prove $AA^\dagger = B^\dagger B + 3$ and $BB^\dagger + 3 = -\partial_x^2 + 4$. Use these identities to classify all L^2 eigenvalues of H_K, including possible embedded or threshold eigenvalues. Check the domains when intertwining.

(b) Normalise its two bound states. Exhibit a bounded solution at spectral value four and explain why it is not another L^2 eigenstate. Using the given theorem, state the full spectrum.

Hint. An eigenfunction is smoother than a generic H^2 vector: bootstrap its second-order equation using the bounded derivatives of the potential. A positive-energy eigenfunction maps under A, then under B, to a free L^2 eigenfunction unless the second map vanishes.

Answer.

$$\sigma(H_K) = \{0, 3\} \cup [4, \infty), \qquad \psi_0 = \frac{\sqrt{3}}{2}\operatorname{sech}^2 x, \qquad \psi_3 = \sqrt{\frac{3}{2}}\operatorname{sech} x \tanh x.$$

Both bound eigenvalues are simple. The threshold solution $(3\tanh^2 x - 1)/2$ is bounded but not in L^2.

Exercise 18.2 (A moving kink and the sign of momentum). For $|v_{\mathrm{tr}}| < 1$, set $\gamma = (1 - v_{\mathrm{tr}}^2)^{-1/2}$ and

$$\phi(t, x) = \tanh\bigl(\gamma(x - v_{\mathrm{tr}} t - x_0)\bigr).$$

Verify the nonlinear field equation directly. At fixed time compute the charge, full energy and the lower-index translation integral $P_1 = \int \phi_t \phi_x \mathrm{d}x$. Find the physical raised momentum P^1 and the relation between its square and the energy square. Check the result at $v_{\mathrm{tr}} = 3/5$. Justify membership in the affine class $\phi - b \in H^1$, $\phi_t \in L^2$.

Hint. Use the static identity $b'' = V'(b)$, then change variable to the boosted coordinate in each integral. Raising a spatial momentum index changes its sign.

Answer.

$$Q = 1, \qquad E = \frac{4}{3}\gamma, \qquad P_1 = -\frac{4}{3}\gamma v_{\mathrm{tr}}, \qquad P^1 = -P_1, \qquad E^2 - (P^1)^2 = \frac{16}{9}.$$

At speed $3/5$, $E = 5/3$ and $P_1 = -1$.

Exercise 18.3 (Removing the linear translation drift). Use the spectral theorem for the self-adjoint operator in Problem 1 as an additional analytic input: on the orthogonal complement of its zero mode, the form-domain inequality is $\mathfrak{q}[h] \geq 3\|h\|_2^2$ for $h \in H^1$ with $(\psi_0, h) = 0$. Prove $\|h\|_{H^1}^2 \leq 2\mathfrak{q}[h]$ on this subspace. For any real solution of $h_{tt} + H_K h = 0$ with

$$h \in C(H^2) \cap C^1(H^1) \cap C^2(L^2),$$

prove that $(\psi_0, h(0)) = (\psi_0, h_t(0)) = 0$ preserves this orthogonality, and deduce a uniform bound on $\|h(t)\|_{H^1}^2 + \|h_t(t)\|_2^2$ from its initial quadratic energy. No existence theorem for this new linear equation is required: the assertion is conditional on a solution in the displayed class.

Hint. The negative part of $4 - 6\,\mathrm{sech}^2 x$ is at most two. Pair the linear equation with ψ_0 and then differentiate its quadratic energy.

Answer. The complete squared norm is bounded by $4\mathcal{E}_{\mathrm{lin}}(0)$, where $\mathcal{E}_{\mathrm{lin}} = (\|h_t\|_2^2 + \mathfrak{q}[h])/2$. This is a linear bound after both translation projections are removed.

Exercise 18.4 (The nonlinear energy remainder on a fixed slice). For real $h \in H^1(\mathbb{R})$, compute the exact remainder in $E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] - \mathfrak{q}[h]/2$. Prove a bound using only $\|h\|_{H^1}$. If $(\psi_0, h) = 0$, show that for $\|h\|_{H^1} \leq 1/(16\sqrt{2})$,

$$E_{\mathrm{stat}}[b+h] - E_{\mathrm{stat}}[b] \geq \frac{1}{8}\|h\|_{H^1}^2.$$

Why is this static estimate insufficient by itself to prove nonlinear orbital stability of the kink?

Hint. Expand the actual quartic potential. Use the stationary equation to remove the linear term, extending the integration by parts by H^1 density. Then combine the embedding constant of Lecture 15 with Problem 3.

Answer.

$$\mathcal{R}[h] = \int (2bh^3 + \tfrac{1}{2}h^4)\mathrm{d}x, \qquad |\mathcal{R}[h]| \leq \sqrt{2}\|h\|_{H^1}^3 + \frac{1}{4}\|h\|_{H^1}^4.$$

The imposed static orthogonality is not automatically preserved by the nonlinear flow.

Classical Field Theory — Exercise Sheet 19

Adjoint identities, matter and surface currents

Hints and answers

All fields and coefficients on this sheet are smooth. Variations and gauge parameters in action identities have compact support inside the coordinate region. Use the Euler sign $\delta S = \int E_a \delta\phi^a$ and the connection conventions of Lecture 19. No equations of motion are imposed unless explicitly stated.

Exercise 19.1 (A second-order adjoint and the elementary converse). On an open coordinate patch $U \subset \mathbb{R}^{1+1}$ with coordinates (t, x), let a scalar parameter $\varepsilon \in C_c^\infty(U)$ produce two variations

$$R(\varepsilon)^1 = (1+tx)\varepsilon_{tt} + 2(t+x^2)\varepsilon_{tx} + e^t\varepsilon_x, \qquad R(\varepsilon)^2 = (1+t^2)\varepsilon_{xx} + x\varepsilon.$$

Compute $R^*(E_1, E_2)$, first in divergence form and then with every coefficient derivative expanded. Prove the compact pairing identity. Suppose an action has the compact first variation $\int E_a[\phi]h^a$ and $R^*_\phi E[\phi] = 0$ on U for every smooth field. Prove that $R_\phi(\varepsilon)$ is an infinitesimal gauge symmetry. Explain why the same converse holds for any finite-order operator $R_\phi(\varepsilon)^a = \sum_I r^{aI}_\alpha[\phi]\partial_I\varepsilon^\alpha$, without interpreting it as a converse for arbitrary conserved currents.

Hint. Integrate the mixed derivative twice, keeping both coefficient derivatives. The converse uses the same pairing in reverse; it requires an identity at every field, not only at solutions.

Answer.

$$R^*E = (1+tx)E_{1,tt} + 2(t+x^2)E_{1,tx} + 6xE_{1,t} + (2-e^t)E_{1,x} + (1+t^2)E_{2,xx} + xE_2.$$

The identity $R^*E = 0$ is sufficient for compact infinitesimal invariance.

Exercise 19.2 (The Euler term carried by charged matter). Write $A_{\mathrm{em}} = A_\mu \mathrm{d}x^\mu$, $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ and let $u = (u_1, u_2)^{\mathsf{T}}$ be two real scalar fields. Given $q, \lambda, v > 0$, set

$$T = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad D_\mu u = \partial_\mu u - qA_\mu Tu, \qquad L = -\frac{1}{4}F_{\mu\nu}F^{\mu\nu} + \frac{1}{2}D_\mu u \cdot D^\mu u - \frac{\lambda}{8}(|u|^2 - v^2)^2.$$

The gauge variations are $\delta A_\mu = \partial_\mu\varepsilon$ and $\delta u = q\varepsilon Tu$.

(a) Verify covariance of $D_\mu u$, derive both Euler arrays by compact variation, and derive the off-shell Noether identity containing them.

(b) On a coordinate patch take $A_\mu = 0$ and $u = (t, t^2)^{\mathsf{T}}$. Compute the ordinary divergence of the total gauge Euler array and its balancing scalar Euler term. Does the pure Maxwell identity remain an identity for that total array by itself?

(c) For a real orthogonal representation with skew generators t_a, use instead the supplied variations $\delta A_\mu = D_\mu\varepsilon$, $\delta u = -\varepsilon^a t_a u$. Assume these variations are a gauge symmetry for every compact parameter. If the compact first variation is $\int [\langle E^\mu_A, \delta A_\mu\rangle + E_u \cdot \delta u]$, derive its covariant component identity. Here A is a connection as in Lecture 19, and the Lie algebra basis is orthonormal in its invariant pairing.

Hint. The two derivatives of the gauge parameter cancel in the covariant matter variation. In the first variation, integrate only the derivative acting on the arbitrary parameter. A radial potential has gradient orthogonal to Tu.

Answer.

$$-\partial_\mu E_A^\mu + qE_u \cdot Tu = 0.$$

In the polynomial example both $\partial_\mu E_A^\mu$ and $qE_u \cdot Tu$ equal $-2qt$. For the connection convention of part (c), the identity is $(D_\mu E_A^\mu)^a + E_u \cdot t_a u = 0$.

Exercise 19.3 (Noncommuting constant connections). Let $T_a = -\mathrm{i}\sigma_a/2$, so $[T_a, T_b] = \epsilon_{abc}T_c$. In pure Yang–Mills theory with coupling $g > 0$, take first $A_1 = T_1$, $A_2 = T_2$, $A_0 = A_3 = 0$, all constant. Compute every nonzero curvature and Euler component $E^\nu = g^{-2}D_\mu F^{\mu\nu}$, and check the covariant identity. Repeat after changing A_3 to $T_1 + T_2$. Exhibit the separate nonzero commutator contributions to $D_\nu E^\nu$ in that second case. What can these constant examples establish about an ordinary-divergence identity for arbitrary nonconstant connections?

Hint. Two raised spatial indices give no metric sign. For constant connections $E^j = g^{-2}\sum_i [A_i, [A_i, A_j]]$; compute each Lie bracket before summing.

Answer. Initially $F_{12} = T_3$ and $(E^1, E^2) = g^{-2}(-T_1, -T_2)$. After the change,

$$(E^1, E^2, E^3) = g^{-2}(-2T_1 + T_2,\ T_1 - 2T_2,\ -T_1 - T_2),$$

with divergence commutators $g^{-2}(T_3, -T_3, 0)$.

Exercise 19.4 (A local superpotential theorem and a Maxwell verification). For this exercise, assume the following local decomposition statement. It is an additional hypothesis here, not a theorem to infer from the fundamental lemma alone. For a finite-order local Lagrangian of unconstrained $\mathbb{R}^N$-valued fields on a star-shaped coordinate patch of spacetime dimension at least two, a finite-order infinitesimal gauge symmetry has a local Noether current of the form

$$j^{\mu}_{\varepsilon} = W^{\mu}_{\varepsilon} + \partial_{\nu} U^{\nu\mu}_{\varepsilon}, \qquad U^{\nu\mu}_{\varepsilon} = -U^{\mu\nu}_{\varepsilon}.$$

The current and these expressions depend linearly on the parameter and finitely many of its derivatives, with smooth finite-order local field coefficients. The remainder W vanishes when the Euler expressions and their derivatives vanish. This is a local statement on the full jet space; no global topological superpotential theorem is supplied.

Verify it directly for

$$L = -\frac{1}{4} F_{\mu\nu} F^{\mu\nu} + A_{\mu} j^{\mu}_{\mathrm{ext}}, \qquad \partial_{\mu} j^{\mu}_{\mathrm{ext}} = 0, \qquad \delta A_{\mu} = \partial_{\mu} \varepsilon,$$

where the external current is fixed and smooth. Derive the gauge Noether current, its Euler remainder and an explicit U. On solutions, compare its spatial integral for compact spatial parameters with a parameter equal to a constant ε_{∞} outside a ball. For the latter comparison assume a smooth Maxwell field whose external current has compact spatial support on the slice and total charge $Q = \int j^{0}_{\mathrm{ext}} \mathrm{d}^3 x$. Use $F_{i0} = E_i$ and an outward sphere orientation to find the surface sign. No Schwartz or L^2 condition on the potential is imposed in this boundary comparison.

Hint. Here the density changes by the divergence of $\varepsilon j^{\mu}_{\mathrm{ext}}$. Keep this term in the first Noether current before integrating the parameter derivative off the curvature.

Answer.

$$j^{\mu}_{\varepsilon} = -F^{\mu\nu} \partial_{\nu} \varepsilon - \varepsilon j^{\mu}_{\mathrm{ext}} = -\varepsilon E^{\mu}_{A} + \partial_{\nu} (\varepsilon F^{\nu\mu}).$$

The on-shell integral is zero for compact spatial parameters and $-\varepsilon_{\infty} Q$ for the stated constant boundary value.

Classical Field Theory — Exercise Sheet 20

Nonabelian constraints and local reduction

Hints and answers

Use $T_a = -\mathrm{i}\sigma_a/2$ for $\mathfrak{su}(2)$, so $[T_a, T_b] = \epsilon_{abc}T_c$ and $\langle T_a, T_b\rangle = \delta_{ab}$. All spatial fields are smooth on $\mathbb{T}^3$. Write $\mathcal{V}_{\mathrm{per}} = (2\pi)^3$ for its volume. The canonical bracket has the order fixed in Lectures 7 and 20.

Exercise 20.1 (A homogeneous constraint manifold). Restrict the spatial fields to constants $A_i = a_i^a T_a$ and $\Pi_i = p_i^a T_a$. A parameter $\varepsilon = \varepsilon^a T_a$ is also constant.

(a) Pull the canonical symplectic form back to these constants and derive their finite-dimensional bracket. Compute the generator $C[\varepsilon] = \mathcal{V}_{\mathrm{per}} \sum_i \langle \Pi_i, [A_i, \varepsilon]\rangle$ and its Hamiltonian vector field. Prove its constraint bracket, including the volume factor and sign.

(b) At $A_i = T_i$ for $i = 1, 2, 3$, write $\Pi_i = \sum_j M_{ij}T_j$ for a real matrix M. Classify all M satisfying $\sum_i [A_i, \Pi_i] = 0$. Give one nonscalar matrix in the constraint set and one outside it. Explain why checking a single homogeneous state cannot prove the constraint algebra on the full periodic field space.

Hint. A homogeneous field variation is constant, so its spatial integral contributes the volume to the symplectic form. Pair the two terms involving M_{ij} and M_{ji} in each Gauss component.

Answer. The bracket is $\mathcal{V}_{\mathrm{per}}^{-1} \sum_{i,a} (F_{a_i^a} G_{p_i^a} - F_{p_i^a} G_{a_i^a})$. The generator gives $\delta A_i = [A_i, \varepsilon]$ and $\delta \Pi_i = [\Pi_i, \varepsilon]$, with $\{C[\varepsilon], C[\zeta]\} = -C[[\varepsilon, \zeta]]$. At $A_i = T_i$, Gauss holds exactly when M is symmetric.

Exercise 20.2 (Integrating a residual gauge generator). For a fixed smooth periodic $\varepsilon(x)$, let $h_s(x) = \exp(s\varepsilon(x))$ and define

$$A_i(s) = h_s^{-1} A_i h_s + h_s^{-1} \partial_i h_s, \qquad \Pi_i(s) = h_s^{-1} \Pi_i h_s.$$

Here s is the transformation parameter, not physical time.

(a) Prove that these arrays solve $\partial_s A_i(s) = D_i^{(s)} \varepsilon$ and $\partial_s \Pi_i(s) = [\Pi_i(s), \varepsilon]$. Show that the transformation preserves the canonical form, the temporal-gauge Hamiltonian and the Gauss constraint surface.

(b) Take $\varepsilon = \sin x^1 \, T_3$, $A_1 = T_1$, $A_2 = T_3$, $A_3 = 0$, $\Pi_1 = T_2$ and $\Pi_2 = \Pi_3 = 0$. Compute the transformed fields and their Gauss array. Is this example on the constraint surface? What changes if a finite gauge transformation depends on physical time?

Hint. Differentiate $h_s^{-1} h_s = 1$ and use $h_s \varepsilon = \varepsilon h_s$. For the constraint use $D_i^{(s)}(h_s^{-1} X h_s) = h_s^{-1}(D_i X) h_s$. Check the derivative part of the transformed Gauss array as well as its commutator.

Answer. The flow is generated by $C[\varepsilon]$ and preserves the stated structures. With $\alpha = s \sin x^1$,

$$A_1(s) = \cos\alpha \, T_1 - \sin\alpha \, T_2 + s \cos x^1 \, T_3, \quad A_2(s) = T_3,$$
$$\Pi_1(s) = \sin\alpha \, T_1 + \cos\alpha \, T_2, \qquad \sum_i D_i^{(s)} \Pi_i(s) = T_3.$$

The example violates Gauss. Temporal gauge is preserved by a finite transformation exactly when its physical-time derivative is zero.

Exercise 20.3 (A linearized reduced bracket and its limit). Fix a real constant η and the background $A_1^0 = \eta T_3$, $A_2^0 = A_3^0 = 0$, $\Pi^0 = 0$. Work only with smooth variations a_i, p_i and parameters having zero spatial mean. This excludes the entire zero Fourier mode from this problem. Put $D_i^0 = \partial_i + [A_i^0, \cdot\,]$.

(a) Derive the linearized Gauss equation $D_i^0 p_i = 0$ and the gauge variation $a_i \mapsto a_i + D_i^0 \varepsilon$. For the Coulomb condition $\partial_i a_i = 0$, compute the eigenvalues of $M = -\partial_i D_i^0$ on each nonzero Fourier mode, using $T_+ = T_1 + \mathrm{i}T_2$, $T_- = T_1 - \mathrm{i}T_2$ and T_3.

(b) For $|\eta| < 1$, prove that M^{-1} maps smooth zero-mean arrays to themselves. Define

$$P = 1 + D^0 M^{-1}\,\mathrm{div}, \qquad P^* = 1 + \nabla M^{-1} D^0 \cdot\,.$$

Prove that Pa is divergence-free and gauge invariant, while $P^* p$ satisfies linearized Gauss. Prove $P^2 = P$. For real smooth zero-mean vector smearings f, g, compute the canonical bracket of $\mathcal{A}[f] = \int \langle f, Pa\rangle$ and $\mathcal{P}[g] = \int \langle g, P^* p\rangle$.

(c) At $\eta = 2$, find a real zero-mean parameter with $M\varepsilon = 0$ but $D^0\varepsilon \neq 0$. Explain why this prevents using the preceding inverse there. Does the calculation establish a global nonlinear reduced bracket?

Hint. The adjoint eigenvalues of T_3 on T_+ and T_- are $-\mathrm{i}$ and i. Use periodic integration by parts for P^*. For the last part try a parameter depending on $x^1 + x^2$.

Answer. At $n \in \mathbb{Z}^3 \setminus \{0\}$, the three eigenvalues are $|n|^2 - \eta n_1$, $|n|^2 + \eta n_1$ and $|n|^2$. For $|\eta| < 1$,

$$\{\mathcal{A}[f], \mathcal{P}[g]\} = \int_{\mathbb{T}^3} \langle f, Pg\rangle \,\mathrm{d}^3x.$$

At $\eta = 2$ one choice is $\varepsilon = T_1 \cos(x^1 + x^2) - T_2 \sin(x^1 + x^2)$. The result is a linear reduction in the specified Fourier sector, not a global nonlinear gauge construction.

Classical Field Theory — Exercise Sheet 21

Vector mass matrices and longitudinal energy

Hints and answers

Use the signature $(+,-,-,-)$ and canonically normalised real scalar kinetic coefficient $1/2$. The questions concern quadratic theories near a fixed vacuum. They make no assertion about a global unitary gauge.

Exercise 21.1 (An orbit determines a vector mass matrix). Let a compact Lie group act orthogonally on $\mathbb{R}^N$, and choose a basis of its Lie algebra, represented by skew real generators T_a, $1 \le a \le r$. Let u_0 be a vacuum of an invariant C^2 potential. Fix $g > 0$, and give the scalar the kinetic term

$$\frac{1}{2}(D_\mu\phi)\cdot(D^\mu\phi), \qquad D_\mu\phi = \partial_\mu\phi + gA^a_\mu T_a\phi.$$

The vector kinetic terms are canonical at the quadratic level.

(a) Derive the vector squared-mass matrix at $\phi = u_0$. Prove that it is nonnegative and identify its kernel and rank. Include generators which act trivially in the representation.

(b) For the three-dimensional rotation representation, take

$$T_a w = e_a \times w, \qquad u_0 = (v/\sqrt{2}, v/\sqrt{2}, 0), \qquad v > 0.$$

Compute the full matrix, its orthonormal eigenvectors and the vector masses. Which gauge-field combination remains massless?

Hint. At a constant vacuum, keep the part of $D_\mu\phi$ linear in A. Evaluate its Gram quadratic form on a real parameter array. For the example use $(a\times u)\cdot(b\times u) = (a\cdot b)|u|^2 - (a\cdot u)(b\cdot u)$.

Answer. The squared-mass matrix is $\mathsf{M}_{ab} = g^2(T_a u_0)\cdot(T_b u_0)$. Its kernel is $\{z : \sum_a z^a T_a u_0 = 0\}$ and its rank is $\dim G - \dim H$ for the vacuum stabiliser H. In the example it is

$$g^2v^2\begin{pmatrix} 1/2 & -1/2 & 0 \\ -1/2 & 1/2 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The combination $(A^1 + A^2)/\sqrt{2}$ is massless. The other two vector masses are gv.

Exercise 21.2 (The charged and neutral electroweak combinations). The following bosonic representation data are given. A complex doublet $\Phi \in \mathbb{C}^2$ has kinetic term $\frac{1}{2}(D_\mu\Phi)^\dagger D^\mu\Phi$, potential $\lambda(\Phi^\dagger\Phi - v^2)^2/8$, and vacuum $\Phi_0 = (0, v)^{\mathsf{T}}$. Here $g, g', v, \lambda > 0$ and

$$D_\mu = \partial_\mu - \mathrm{i}\left(gW_\mu^a\frac{\sigma_a}{2} + g'B_\mu\frac{I_2}{2}\right).$$

All four real vector fields have canonical quadratic Maxwell kinetic terms. The Pauli matrices and the hypercharge generator $Y = I_2/2$ specify the representation; no fermion sector is included.

(a) Find the full real squared-mass matrix in the order (W^1, W^2, W^3, B). Diagonalise its neutral block and define Z and the massless photon A_γ with unit kinetic normalisation. Define $W^\pm = (W^1 \mp \mathrm{i}W^2)/\sqrt{2}$ and find their common mass.

(b) Find the generator annihilating Φ_0 and the coefficient of its photon coupling in D_μ. Find the radial scalar mass and compare the quadratic number of physical modes in the uncoupled limit $g, g' \to 0$ with the broken quadratic description at positive couplings. Explain why W^+ and W^- do not count as two independent complex vectors.

This is the bosonic mass calculation of the electroweak model. For the original construction see Weinberg, *Physical Review Letters* **19** (1967), 1264–1266, equations (4), (7)–(12). The signs and scalar normalisation to use are those explicitly given above.

Hint. Apply the covariant derivative to the two entries of Φ_0 before squaring. The neutral scalar combination is $-gW^3 + g'B$. Diagonalise that rank-one real quadratic form by an orthogonal rotation.

Answer. Writing $g_Z = \sqrt{g^2 + g'^2}$,

$$Z = \frac{gW^3 - g'B}{g_Z}, \qquad A_\gamma = \frac{g'W^3 + gB}{g_Z}, \qquad m_W = \frac{gv}{2}, \quad m_Z = \frac{g_Z v}{2}, \quad m_\gamma = 0.$$

The unbroken generator is $\sigma_3/2 + Y$, with photon coupling $e = gg'/g_Z$. The radial mass is $\sqrt{\lambda}\,v$. Both descriptions contain twelve physical quadratic modes.

Exercise 21.3 (The positive reduced Proca Hamiltonian). Use Lecture 21's variables $B = -\beta\,\mathrm{d}t + \mathbf{b}\cdot\mathrm{d}x$, mass $m > 0$ and spatial momentum $\mathbf{\Pi}$. Take real Schwartz data on $\mathbb{R}^3$. Before elimination the Hamiltonian is

$$H_B = \int_{\mathbb{R}^3} \left[\frac{1}{2}|\mathbf{\Pi}|^2 + \frac{1}{2}|\operatorname{curl}\mathbf{b}|^2 + \frac{1}{2}m^2|\mathbf{b}|^2 + \beta \operatorname{div}\mathbf{\Pi} - \frac{1}{2}m^2\beta^2 \right] \mathrm{d}^3x.$$

Its scalar constraints are $\pi_\beta = 0$ and $\operatorname{div}\mathbf{\Pi} - m^2\beta = 0$.

(a) Eliminate the scalar pair and derive the reduced Hamiltonian and both spatial Hamilton equations with the remaining canonical form. Prove positivity and show that a zero-energy real datum is zero.

(b) At a nonzero Fourier vector k, split the spatial amplitudes into longitudinal and transverse parts. Find the frequencies and a canonical normalisation of each real longitudinal Fourier quadrature. Treat $k = 0$ separately. Why does the energy formula not have a regular mass-zero limit for fixed arbitrary longitudinal momentum?

Individual Fourier modes here analyse the multiplier; they are not claimed to be finite-energy plane-wave fields on $\mathbb{R}^3$.

Hint. Complete the square in β. For the longitudinal part, div contributes a factor $\mathrm{i}|k|$. To normalise an oscillator, rescale its coordinate and momentum by reciprocal factors.

Answer. The reduced energy and equations are

$$H_{\mathrm{red}} = \frac{1}{2}\int \left[|\mathbf{\Pi}|^2 + |\operatorname{curl}\mathbf{b}|^2 + m^2|\mathbf{b}|^2 + m^{-2}|\operatorname{div}\mathbf{\Pi}|^2 \right] \mathrm{d}^3x,$$

$$\mathbf{b}_t = \mathbf{\Pi} - m^{-2}\nabla \operatorname{div}\mathbf{\Pi}, \qquad \mathbf{\Pi}_t = -\operatorname{curl}\operatorname{curl}\mathbf{b} - m^2\mathbf{b}.$$

Every polarisation has frequency $\nu = \sqrt{|k|^2 + m^2}$. For a real longitudinal quadrature, $Q_L = (m/\nu)b_L$ and $P_L = (\nu/m)\Pi_L$ are canonical.

Classical Field Theory — Exercise Sheet 22

Vortex orientation, scalar reduction and moduli

Hints and answers

Throughout this sheet $q = v = \lambda = 1$, $D_j = \partial_j - \mathrm{i}A_j$, $B = \partial_1 A_2 - \partial_2 A_1$, and the plane has orientation $\mathrm{d}x_1 \wedge \mathrm{d}x_2$. Use Lecture 22's energy and current $K_j = \frac{1}{2}\operatorname{Im}(\overline{\Phi} D_j \Phi)$. Every sector statement retains its finite-energy, uniform vacuum-modulus and absolute magnetic-integrability assumptions.

Exercise 22.1 (Reflection and higher-winding antivortices). Let $R(x_1, x_2) = (x_1, -x_2)$. For smooth fields set $\widetilde{\Phi} = \Phi \circ R$ and $\widetilde{A} = R^* A$.

(a) Track B, K, the energy and counterclockwise winding under this reflection. Derive the square completion for winding $-N$, $N \geq 1$, and prove the bound and equality equations in the reflected sector.

(b) For real radial profiles use $\Phi = f(r)e^{-\mathrm{i}N\theta}$ and $A = \alpha(r)\mathrm{d}\theta$. Derive both equality ODEs, the flux and the boundary conditions on the positive vacuum branch. Give sufficient smooth Cartesian core forms and the first corrections to $f = s_N r^N + \cdots$ and $\alpha = -r^2/4 + \cdots$. Here $s_N > 0$ is a local slope coefficient, not a proved global shooting value.

Hint. Reflection reverses curl and winding but not the energy integral. For the radial calculation use $\partial_1 - \mathrm{i}\partial_2 = e^{-\mathrm{i}\theta}(\partial_r - \mathrm{i}r^{-1}\partial_\theta)$. Integrate the logarithmic derivative of f/r^N near zero.

Answer. The negative-winding identity and equality equations are

$$e_1 = \frac{1}{2}|(D_1 - \mathrm{i}D_2)\Phi|^2 + \frac{1}{2}\left[B + \frac{1 - |\Phi|^2}{2}\right]^2 - \frac{B}{2} - \operatorname{curl} K,$$

$$E_1 \geq \pi N, \qquad (D_1 - \mathrm{i}D_2)\Phi = 0, \quad B = -\frac{1}{2}(1 - |\Phi|^2).$$

The radial equations are $f' = (N + \alpha)f/r$ and $\alpha'/r = -(1 - f^2)/2$, with $\alpha(\infty) = -N$ and flux $-2\pi N$. The leading corrections are $f = s_N r^N(1 - r^2/8 + O(r^4))$ and $\alpha = -r^2/4 + s_N^2 r^{2N+2}/(4N + 4) + O(r^{2N+4})$.

Exercise 22.2 (The scalar equation with its point sources). Let a positive critical vortex in the stated sector have finitely many zeros $z_j \in \mathbb{R}^2$ of multiplicities $m_j \geq 1$. Away from the zeros put $w = \log|\Phi|^2$. Assume as local zero data that $w = 2m_j \log|x - z_j| + w_j$ near z_j, where w_j is smooth. This local regular remainder is a hypothesis, not a claim about arbitrary smooth fields with zeros. Use the distributional identity $\Delta \log|x - z_j| = 2\pi\delta_{z_j}$ as a given.

(a) On a phase patch derive A_1, A_2 in terms of w and the real phase. Eliminate the potential from the two Bogomolny equations.

(b) Prove the resulting equation on the whole plane as a distribution, including its point-source coefficient. Determine $\int_{\mathbb{R}^2}(1 - e^w)\mathrm{d}^2x$ from the sector flux when $\sum_j m_j = N$. Does this reduction prove existence for prescribed zeros?

Hint. Write $\Phi = e^{w/2 + \mathrm{i}\theta}$ on a nonvanishing patch and separate the real and imaginary parts of $(D_1 + \mathrm{i}D_2)\Phi = 0$. The regular remainder lets the nonsingular part of the equation extend through each zero.

Answer. The equation and integral are

$$\Delta w + 1 - e^{w} = 4\pi \sum_j m_j \delta_{z_j}, \qquad \int_{\mathbb{R}^2} (1 - e^{w}) \mathrm{d}^2 x = 4\pi N.$$

They are consequences for the assumed vortex, not an existence proof.

Exercise 22.3 (Parameters at a two-vortex collision). A *moduli space* here is the set of critical positive N-vortex solutions with the vacuum asymptotics, modulo smooth gauge transformations. Use the following stronger geometric result as a *given*: this space is a nonsingular complex manifold, and the coefficients of

$$P(z) = \prod_{j=1}^{N} (z - z_j)$$

are smooth complex coordinates, including at coincident zeros. This input is not inferred from existence and uniqueness as a set alone. For its zero-coordinate description see Alonso Izquierdo, Garcia Fuertes, Manton and Mateos Guilarte, *JHEP* **01** (2024) 020, §5, pp. 10–11, equation (5.1) and discussion. Taubes's existence/uniqueness result is the one cited in Lecture 22.

(a) Count the real moduli and the moduli after fixing the mean zero position. Explain why ordering the zeros overcounts configurations.

(b) For $N = 2$, write $P(z) = z^2 - sz + p$. Replace (s, p) by the centre $Z = s/2$ and discriminant $\Delta = s^2 - 4p$. Locate the collision set and determine its real codimension. Explain why (z_1, z_2) and their separation are poor smooth coordinates there.

Hint. A monic degree-N polynomial has N free complex coefficients. For two zeros complete the square; do not choose a branch of a square root and assume that branch is smooth at zero.

Answer. There are $2N$ real moduli and $2N - 2$ after centring. For two vortices, $P(z) = (z - Z)^2 - \Delta/4$ and the collision set is $\Delta = 0$. It has real codimension two in the four-dimensional moduli space.

Exercise 22.4 (Designing a radial computation). For the positive unit-vortex ODEs $f' = (1-\alpha)f/r$ and $\alpha'/r = (1-f^2)/2$, design a computation on $[r_0, R]$, where $0 < r_0 \ll 1 < R$.

(a) Derive the coefficients through r^5 in f and through r^6 in α, using the free coefficient $s_0 = f'(0)$.

(b) Treat s_0 as an unknown parameter. Give three boundary residuals for a two-equation boundary-value solver using the core series and $f(R) = 1$. Explain why adding $\alpha(R) = 1$ independently would overdetermine this particular setup.

(c) Specify separate tests of origin truncation, outer-boundary truncation and solver accuracy. What further information is needed before a numerical residual could prove existence of a full-plane vortex? No numerical run is required in this problem.

Hint. Substitute odd and even power series in the two ODEs and match coefficients in increasing order. Count the free parameter along with the differential-equation order before choosing endpoint conditions.

Answer. The series are

$$f = s_0 r - \frac{s_0}{8} r^3 + \left(\frac{s_0}{128} + \frac{s_0^3}{32} \right) r^5 + O(r^7), \qquad \alpha = \frac{r^2}{4} - \frac{s_0^2 r^4}{8} + \frac{s_0^2 r^6}{48} + O(r^8).$$

Use both series at r_0 and $f(R) - 1$ as the three residuals. Refinement tests diagnose approximation errors but do not alone prove full-plane existence.

Classical Field Theory — Exercise Sheet 23

A second KdV bracket and further conserved quantities

Hints and answers

Use Lecture 23's field space $\mathcal{S} = \mathcal{S}(\mathbb{R}, \mathbb{R})$, gradient space $\mathcal{B} = \mathbb{R} \oplus \mathcal{S}$ and admissible smooth functionals $\mathcal{A}_{\mathrm{KdV}}$. Write $D_x = \partial_x$ for spatial differentiation. Every time-dependent conclusion is conditional on a C^1 path in $\mathcal{S}$ satisfying KdV.

Exercise 23.1 (A variable Hamiltonian operator). For $u \in \mathcal{S}$ define the linear operator $J_2[u] : \mathcal{B} \to \mathcal{S}$ by

$$J_2[u]b = -b''' + 4ub' + 2u'b.$$

The prime means D_x, including when applied to a gradient. Define $\{F, G\}_2 = \int_{\mathbb{R}} F_u J_2[u] G_u \, \mathrm{d}x$.

(a) Prove skewness by integration by parts with both gradients in $\mathcal{B}$, keeping the endpoint primitive. Derive the gradient of $\{F, G\}_2$ and prove closure of the stated class.

(b) Prove Jacobi, also for $J_2 + \alpha D_x$ with every constant $\alpha \in \mathbb{R}$. Use the Hessian symmetry from Lecture 23 and account for the extra terms from differentiating the operator itself. A useful notation is $W(a, b) = ab' - a'b$.

(c) For $P = \frac{1}{2}\int u^2$ and $H = \int (u^3 + u_x^2/2)$, prove $J_2 P_u = D_x H_u$. Is the Gardner mass Casimir also a Casimir of this bracket?

The second KdV operator is discussed in Olver, *Applications of Lie Groups to Differential Equations*, second edition, Examples 7.6 and 7.9, pp. 438–439 and 444. The operator above states the course's scaling and signs explicitly; its proof is part of this problem.

Hint. For the operator derivative use $D(J_2[\cdot]b)(u)h = 4hb' + 2h'b$ when b is held fixed. After the Hessian terms cancel, show that the cyclic third-derivative terms form the derivative of $W(a, b)c'' + W(b, c)a'' + W(c, a)b''$.

Answer. Writing $a = F_u$, $b = G_u$ and $H_F = D(F_u)(u)$, the bracket gradient is

$$(\{F, G\}_2)_u = H_F J_2 b - H_G J_2 a + 2(ab' - a'b).$$

The bracket is Poisson on the stated class. Both representations give $6uu' - u'''$. The mass is not a J_2 Casimir because $J_2 1 = 2u'$.

Exercise 23.2 (Finding a higher polynomial invariant). For real constants A, B, C put

$$I_{A,B,C}[u] = \int_{\mathbb{R}} \left(Au^4 + Bu(u')^2 + C(u'')^2 \right) \mathrm{d}x.$$

Determine all triples such that $DI_{A,B,C}(u)(6uu' - u''') = 0$ for every $u \in \mathcal{S}$. Conclude conservation along every stated Schwartz KdV path. Show necessity as well as sufficiency; do not infer independence of two integrals merely by looking at different integrands.

Hint. Compute the Euler derivative first. Reduce the resulting derivative to integrals of $u(u')^3$ and $u'(u'')^2$. To separate them use amplitude rescaling and the asymmetric Schwartz profile $(1 + x)e^{-x^2}$.

Answer. All triples are $(A, B, C) = \zeta_4(5, 10, 1)$, with $\zeta_4 \in \mathbb{R}$. In particular, $\int [5u^4 + 10u(u')^2 + (u'')^2] \mathrm{d}x$ is conserved.

Exercise 23.3 (Classifying Gardner Casimirs). A Casimir of Lecture 23's Gardner bracket is an admissible functional commuting with every admissible functional. Prove that these are exactly

$$F[u] = \psi(M[u]), \qquad M[u] = \int_{\mathbb{R}} u \, \mathrm{d}x,$$

for smooth $\psi : \mathbb{R} \to \mathbb{R}$. Constant scalar functionals are included. Explain why this does not say that every local polynomial density is a nonlinear function of the density u. Compare the local-functional discussion in Olver, Example 7.13, p. 446: the present admissible class also allows nonlocal composites $\psi(M)$.

Hint. Bracket with every smeared field $u(g)$, $g \in \mathcal{S}$. Then move the spatial derivative onto the gradient. For the converse and global classification, fix one $\eta \in \mathcal{S}$ with integral one and use the affine mass fibres.

Answer. A Casimir has a gradient constant in x at every field. It is constant on each mass fibre, so $\psi(s) = F[s\eta]$ gives $F = \psi \circ M$. Conversely its gradient is $\psi'(M)$ in $\mathcal{B}$, killed by D_x. This is a classification of functionals, not a pointwise density statement.

Classical Field Theory — Exercise Sheet 24

A higher Lax flow and operator domains

Hints and answers

Use $D_x = \partial_x$ and the ordered commutator $[B, L] = BL - LB$. All operator calculations in the first problem are on $C_c^\infty(\mathbb{R}, \mathbb{C})$, with smooth real coefficient u. A displayed differential expression does not specify its Hilbert-space domain.

Exercise 24.1 (Determining a fifth-order Lax flow). Let $L = -D_x^2 + u$. For real constants $r_1, \dots, r_6$ consider

$$B_5 = 16D_x^5 + r_1 u D_x^3 + r_2 u' D_x^2 + (r_3 u'' + r_4 u^2) D_x + r_5 u''' + r_6 u u'.$$

Products with u and its derivatives here mean multiplication before the indicated derivative of the test.

(a) Determine the constants so that $[B_5, L]$ is multiplication for every smooth u. Find the resulting scalar coefficient and write the corresponding evolution equation $L_t = [B_5, L]$.

(b) Prove that the resulting B_5 is formally skew-symmetric on compact smooth tests. Distinguish this from a statement about a closed skew-adjoint operator on L^2.

Hint. Collect coefficients of test derivatives in descending order, starting with the fourth derivative. Leibniz's rule makes this a finite triangular system. For the formal adjoint reverse each product before moving its derivatives.

Answer. The coefficients are $(-40, -60, -50, 30, -15, 30)$, giving

$$B_5 = 16D_x^5 - 40uD_x^3 - 60u'D_x^2 + (30u^2 - 50u'')D_x + 30uu' - 15u'''.$$

The scalar flow is $u_t = u^{(5)} - 10uu''' - 20u'u'' + 30u^2u'$. The operator is formally skew-symmetric on the stated test domain.

Exercise 24.2 (Constancy for a sufficiently regular eigenbranch). Let $u \in C^1(I; \mathcal{S}(\mathbb{R}, \mathbb{R}))$ solve $u_t = K_5[u]$ on an open interval I. Suppose $\lambda \in C^1(I; \mathbb{R})$ and $g \in C^1(I; H^7(\mathbb{R}, \mathbb{C}))$, with $g(t) \neq 0$ and $L(t)g(t) = \lambda(t)g(t)$ at every time. The Sobolev space H^7 uses the Fourier norm $\|\langle k\rangle^7 \widehat{g}\|_2$.

Use the following operator fact as a *given*: a real bounded multiplication perturbation of $-D_x^2$ is self-adjoint on the common domain $H^2(\mathbb{R})$. This is the relative-bound-zero case of Kato, *Perturbation Theory for Linear Operators*, V, Theorem 4.3, p. 287. Prove that λ is constant. Specify the domain needed for moving L across an inner product. Does this calculation require simplicity or isolation, and does it construct a branch or prove a theorem about the whole spectrum?

Hint. Differentiate the eigenvalue equation and pair with g. The coefficient identity from the first problem applies to the actual smooth eigenfunction once both ordered compositions belong to L^2.

Answer. The branch satisfies $\lambda' \|g\|_2^2 = (g, [B_5, L]g) = 0$. Here $B_5 g \in H^2$, so self-adjointness is applicable. No simplicity or isolation is required for this conditional computation. It does not supply eigenbranches or constancy of the entire spectrum.

Exercise 24.3 (The same interior expression with moving endpoints). Fix $\ell > 0$ and a smooth positive nonconstant function $\kappa(t)$. On $L^2(0,\ell)$ consider the expression $L = -\partial_x^2$ with domain

$$\mathcal{D}_t = \left\{ f \in H^2(0,\ell) : f(0) = 0, \quad f'(\ell) = \beta(t) f(\ell) \right\}, \qquad \beta(t) = \kappa(t)\coth(\kappa(t)\ell).$$

The trace values in this formula have their usual continuous Sobolev representatives on a finite interval.

(a) Show that this realization is symmetric by its endpoint Green formula. Find a nonzero smooth eigenfunction with a nonconstant eigenvalue. Self-adjointness is not required for this task.

(b) With $B = 0$, verify $L_t\psi = [B, L]\psi = 0$ on every compactly supported interior test. Explain why this does not contradict the preceding conditional eigenbranch result or establish constancy of the varying eigenvalue.

Hint. Try $\sinh(\kappa(t)x)$. The endpoint formula contains values of both functions and their first derivatives, not just the interior differential coefficients.

Answer. The eigenfunction is $\sinh(\kappa(t)x)$ with eigenvalue $-\kappa(t)^2$. The interior test identity is zero, but the domains $\mathcal{D}_t$ vary. The moving endpoint condition prevents using a single self-adjoint domain to cancel the differentiated-eigenfunction terms.

Classical Field Theory — Exercise Sheet 25

Analytic extension and direct scattering

Hints and answers

Use $L = -\partial_x^2 + u$, the Wronskian $W(f,g) = fg' - f'g$, and $f_- = af_+(\cdot,-k) + bf_+(\cdot,k)$. For real $k \neq 0$, the right reflection amplitude is $r = b/a$. All potentials in this sheet are real Schwartz functions.

Exercise 25.1 (An upper-half-plane Volterra construction). For $\operatorname{Im} k \geq 0$ define

$$H(s,k) = \int_0^s e^{2ik\sigma} d\sigma, \qquad s \geq 0.$$

Thus $H(s,k) = (e^{2iks} - 1)/(2ik)$ for $k \neq 0$ and $H(s,0) = s$. Consider

$$m(x,k) = 1 + \int_x^\infty H(y-x,k)u(y)m(y,k)dy.$$

(a) For every finite R, construct the unique bounded solution on $x \geq R$ by iteration, uniformly for k in the closed upper half-plane. Prove that $m(x,k)$ is holomorphic in $\operatorname{Im} k > 0$ for each fixed x, and continuous up to its real boundary, including $k = 0$.

(b) Prove $m \to 1$ and $m_x \to 0$ as $x \to +\infty$. Show that $f_+(x,k) = e^{ikx}m(x,k)$ solves $Lf_+ = k^2 f_+$. State the two right asymptotic conditions at zero energy. Does this prove a nonzero Wronskian basis at $k = 0$?

The complex-analysis inputs may be used as givens: dominated parameter integrals of holomorphic functions are holomorphic, and locally uniform limits of holomorphic functions are holomorphic. These are the parameter-integral theorem and the uniform-limit result proved in *Analysis*, Lecture 10.

Hint. For $\operatorname{Im} k \geq 0$, $|H(s,k)| \leq s$. On $x \geq R$, bound each factor $y - x$ by $y - R$ and use ordered-simplex integration. The weighted tail $\int_R^\infty (y-R)|u(y)|dy$ is finite.

Answer. With $W_R = \int_R^\infty (y-R)|u(y)|dy$, the jth iterate is bounded by $W_R^j/j!$. The solution is holomorphic in the open upper half-plane and continuous on its closure. At $k = 0$, $f_+ \to 1$ and $f_+' \to 0$. The Wronskian formula $2ik$ gives no basis at zero.

Exercise 25.2 (Reflectionless scattering by intertwining). Fix $\kappa > 0$ and set

$$L_\kappa = -\partial_x^2 - 2\kappa^2 \operatorname{sech}^2(\kappa x), \qquad A = \partial_x + \kappa \tanh(\kappa x), \quad A^\dagger = -\partial_x + \kappa \tanh(\kappa x).$$

(a) Factor $L_\kappa + \kappa^2$ and the free operator plus κ^2. Use the resulting intertwining identity to construct both real-energy Jost solutions and determine a, b, r with the fixed Wronskian convention.

(b) Find the right-normalised negative-energy state and its inverse squared norm. Show that there are no other negative-energy L^2 states. Then give a bounded zero-energy solution which is not in L^2.

Use the Sobolev integration-by-parts and domain arguments of Lecture 18. A bounded non-L^2 zero-energy solution is called a *zero-energy resonance* in this problem.

Hint. Apply $A^\dagger$ to a free exponential and divide by its right limiting coefficient. The free negative-energy equation has no nonzero L^2 solution, so applying A to any putative bound state is restrictive.

Answer. For real $k \neq 0$,

$$f_+ = e^{ikx}\frac{k + i\kappa\tanh(\kappa x)}{k + i\kappa},$$
$$f_- = e^{-ikx}\frac{k - i\kappa\tanh(\kappa x)}{k + i\kappa},$$
$$a = \frac{k - i\kappa}{k + i\kappa}, \qquad b = r = 0.$$

The unique negative energy is $-\kappa^2$, $f_+(x, i\kappa) = \frac{1}{2}\operatorname{sech}(\kappa x)$ and its norming constant is 2κ. The zero resonance is $\tanh(\kappa x)$.

Exercise 25.3 (A weak asymmetric potential retains a phase). Let $u_\epsilon = \epsilon V$, with $\epsilon \in \mathbb{R}$, and

$$V(x) = e^{-(x-1)^2} + 2e^{-(x+2)^2}.$$

For each fixed real $k \neq 0$, derive the first-order expansions of a, b, r as $\epsilon \to 0$ and prove an $O(\epsilon^2)$ remainder bound. Evaluate the Gaussian coefficient and the leading reflection probability. Show that no negative-energy bound states occur when $\epsilon \geq 0$. The expansion is a fixed-k assertion; no uniform limit at $k = 0$ is requested.

Hint. Use the left Volterra equation and separate its sine into two exponentials. The uniform estimate for $f_- - e^{-ikx}$ is exponential in $|\epsilon|\|V\|_1/|k|$ minus one. Retain the complex Fourier integral before taking its absolute square.

Answer. The expansions are

$$a = 1 - \frac{3\epsilon\sqrt{\pi}}{2ik} + O(\epsilon^2),$$
$$b = \epsilon b_1 + O(\epsilon^2), \quad r = \epsilon b_1 + O(\epsilon^2),$$
$$b_1 = \frac{\sqrt{\pi}e^{-k^2}}{2ik}(e^{-2ik} + 2e^{4ik}).$$

The probability is $\epsilon^2\pi e^{-2k^2}(5 + 4\cos 6k)/(4k^2) + O(|\epsilon|^3)$.

Classical Field Theory — Exercise Sheet 26

Scattering evolution and soliton shifts

Hints and answers

Use the standing real KdV solution $u \in C^\infty(\mathbb{R}; \mathcal{S})$ of Lecture 24 and the Jost/scattering conventions of Lecture 25. The first problem fills the analytic step left sketched in Lecture 26. The later finite-rank construction prescribes its norming coefficients; it makes no additional theorem about birth of eigenvalues.

Exercise 26.1 (Time differentiation at real energy). Fix real $k \neq 0$. For the right Jost solution put $f_+(t,x,k) = e^{ikx} m(t,x,k)$, with

$$m = 1 + \int_x^\infty \frac{e^{2ik(y-x)} - 1}{2ik}\, u(t,y) m(t,y,k)\, dy.$$

(a) Prove that this Volterra construction is continuously differentiable in t. On every compact time interval prove $m_t \to 0$ as $x \to +\infty$, together with the spatial derivative limits needed to compare both the function and derivative tails of $(f_+)_t - Bf_+$. Here $B = -4\partial_x^3 + 6u\partial_x + 3u_x$.

(b) Prove the right and left temporal equations, justify differentiability of a, b from finite-point Wronskians, and prove $r(k,t) = r(k,t_0) e^{8ik^3(t-t_0)}$. Do not pair the oscillatory Jost solutions in L^2.

Hint. Differentiate each ordered Volterra product once in time. The sum over its possible differentiated factors has a factorial majorant. Obtain higher spatial derivatives from the modified scalar ODE. Jost uniqueness requires both a function tail and a first-derivative tail.

Answer. At real energy,

$$(f_+)_t = Bf_+ - 4ik^3 f_+, \qquad (f_-)_t = Bf_- + 4ik^3 f_-,$$
$$a_t = 0, \qquad b_t = 8ik^3 b.$$

The required tail estimates follow from compact-time L^1 tails of u, u_t and their spatial derivatives.

Exercise 26.2 (A positive finite-rank inverse equation). Take distinct $\kappa_j > 0$ and $c_j > 0$, $1 \le j \le N$, with $F(s) = \sum_j c_j e^{-\kappa_j s}$. Consider the right Marchenko equation

$$K(x,y) + F(x+y) + \int_x^\infty K(x,z) F(z+y)\, dz = 0, \qquad y \ge x.$$

The finite positive reflectionless case of the inverse-scattering theorem is a *given*: these data are admissible and reconstruct a real rapidly decaying potential by $u = -2\, dK(x,x)/dx$. It is the zero-background specialization of the cited theorem in Lectures 25–26: Egorova–Grunert–Teschl, *Nonlinearity* **22** (2009), §4, Theorems 4.3–4.4, equations (4.26)–(4.30). Their $(\gamma_j^+)^2$ is c_j.

(a) Reduce the equation to a real symmetric $N \times N$ linear system. Prove that its matrix is positive definite and invertible for every real x.

(b) With $v_j(x) = \sqrt{c_j} e^{-\kappa_j x}$, define $C_{jl} = v_j v_l/(\kappa_j + \kappa_l)$ and $\tau = \det(I + C)$. Prove $K(x,x) = (\log\tau)'$ and $u = -2(\log\tau)''$. Explain why no real pole arises from these positive data.

Hint. Write $K = -\sum_j z_j(x)e^{-\kappa_j y}$ and divide its coefficient equations by $\sqrt{c_j}$. The matrix C is an integral of a real outer product on $[x,\infty)$. Use the determinant derivative formula.

Answer. Putting $q_j = z_j/\sqrt{c_j}$ gives $(I+\mathsf{C})q = v$ and $K(x,x) = -v^{\mathsf{T}}q$. Since $\mathsf{C}' = -vv^{\mathsf{T}}$, $\tau'/\tau = -v^{\mathsf{T}}(I+\mathsf{C})^{-1}v = K(x,x)$. The matrix $I+\mathsf{C}$ is positive definite, so $\tau > 0$.

Exercise 26.3 (Two pulses and their shifts). Take $\kappa_1 > \kappa_2 > 0$ and prescribe $c_j(t) = c_j(0)e^{8\kappa_j^3 t}$, with $c_j(0) > 0$. Put

$$x_{j,0} = \frac{1}{2\kappa_j}\log\frac{c_j(0)}{2\kappa_j}, \qquad \eta_j = -2\kappa_j(x - 4\kappa_j^2 t - x_{j,0}).$$

(a) Evaluate the two-state determinant and find its interaction coefficient d_{12} in $\tau = 1 + e^{\eta_1} + e^{\eta_2} + d_{12}e^{\eta_1+\eta_2}$.

(b) Verify directly that $u = -2\partial_x^2 \log\tau$ solves KdV. One may use, after deriving it by differentiating $\log\tau$,

$$\tau\tau_{xt} - \tau_x\tau_t + \tau\tau_{xxxx} - 4\tau_x\tau_{xxx} + 3\tau_{xx}^2 = 0$$

as a sufficient condition.

(c) Find the incoming and outgoing centre lines of each separated pulse and its net shift. Justify the two spatial derivatives in each moving-frame limit, rather than compare only pointwise values of τ.

The parameters $x_{j,0}$ describe the norming coefficients. They need not be actual minima positions during the interaction.

Hint. For the PDE check expand the symmetric bilinear expression on pairs of exponentials; only the mixed coefficient is nontrivial. For the shifts, factor out the large exponential in each moving frame. Its logarithm is affine in x and contributes nothing to the second derivative.

Answer. The coefficient and shifts are

$$d_{12} = \left(\frac{\kappa_1 - \kappa_2}{\kappa_1 + \kappa_2}\right)^2,$$

$$\Delta x_1 = -\frac{\log d_{12}}{2\kappa_1} > 0, \qquad \Delta x_2 = \frac{\log d_{12}}{2\kappa_2} < 0.$$

The faster pulse advances and the slower one is delayed.

Classical Field Theory — Exercise Sheet 27

Matrix compatibility, formal charges and Bäcklund kinks

Hints and answers

The first two problems assume a smooth real KdV solution $u : I \times \mathbb{R} \to \mathbb{R}$, with I an open time interval, $u_t = 6uu_x - u_{xxx}$ and spatial period $L_{\mathrm{per}} > 0$. This is a conditional hypothesis, not an existence assertion on the circle. The last problem instead uses a real field on the whole line.

Exercise 27.1 (Converting the scalar KdV pair to matrices). For fixed $k \in \mathbb{C} \setminus \{0\}$ consider

$$-\psi_{xx} + u\psi = k^2\psi, \qquad \psi_t = -4\psi_{xxx} + 6u\psi_x + 3u_x\psi.$$

Set $\Psi = (\psi, \psi_x)^{\mathsf{T}}$.

(a) Eliminate higher spatial derivatives to derive $\Psi_x = U\Psi$, $\Psi_t = V\Psi$. Compute every entry of $U_t - V_x + [U, V]$ for an arbitrary smooth u.

(b) For the assumed periodic KdV field, state exactly what Lecture 27's monodromy theorem proves for these matrices. Does it establish the Poisson commutativity or independence of the resulting traces?

Hint. Differentiate $\psi_{xx} = (u - k^2)\psi$ once before substituting into the temporal equation. Differentiate the resulting temporal first row once to obtain the second row.

Answer. The pair and its curvature are

$$U = \begin{pmatrix} 0 & 1 \\ u - k^2 & 0 \end{pmatrix}, \qquad V = \begin{pmatrix} -u_x & 2u + 4k^2 \\ -u_{xx} + (2u + 4k^2)(u - k^2) & u_x \end{pmatrix},$$

$$U_t - V_x + [U, V] = \begin{pmatrix} 0 & 0 \\ u_t - 6uu_x + u_{xxx} & 0 \end{pmatrix}.$$

Every trace power of its period transport is conserved. Independence and involution are not established by that result.

Exercise 27.2 (Local charges from a formal Riccati series). On a patch with $\psi \neq 0$, the scalar equation suggests $\rho = \psi_x/\psi$ and $\rho_x + \rho^2 = u - k^2$. In this problem work purely with formal Laurent series: put $\zeta = 2ik$ and seek

$$\rho = \frac{\zeta}{2} + \sum_{j \geq 1} a_j[u]\zeta^{-j}.$$

Equality means equality of each coefficient; convergence and a globally nonvanishing actual ψ are not assumed.

(a) Find the recursion and the first five coefficients. Integrate them over one spatial period and identify the first three nonzero integrals.

(b) Prove the formal identity $\rho_t = \partial_x[-u_x + (2u + 4k^2)\rho]$ under KdV. Conclude coefficientwise conservation. Explain which analytic conclusion about an actual spectral expansion has not been proved.

Hint. Subtract the leading $\zeta/2$ in the Riccati equation. For the second part set $S = -u_x + (2u + 4k^2)\rho$ and apply $\partial_x + 2\rho$ to $\rho_t - S_x$. A nonzero formal Laurent series bounded above in powers of ζ cannot be annihilated by $\partial_x + \zeta + O(\zeta^{-1})$.

Answer. The recursion is $a_1 = u$ and $a_{n+1} = -a_n' - \sum_{i+j=n,\ i,j\geq 1} a_i a_j$. The nonzero integrated coefficients through order five are

$$\int a_1 = \int u, \qquad \int a_3 = -\int u^2, \qquad \int a_5 = \int [(u')^2 + 2u^3].$$

They are conserved as formal coefficients; convergence or an analytic monodromy asymptotic theorem is not claimed.

Exercise 27.3 (Bäcklund transformation and a sine-Gordon kink). On $\mathbb{R}^{1,1}$ use $\xi = (x+t)/2$, $\eta = (x-t)/2$. For two real C^2 fields $\phi, \widetilde{\phi}$, put $\alpha_B = (\phi + \widetilde{\phi})/2$ and $\beta_B = (\phi - \widetilde{\phi})/2$. For a fixed $a_B > 0$, the proposed *Bäcklund system* is

$$\partial_\xi \alpha_B = a_B \sin \beta_B, \qquad \partial_\eta \beta_B = a_B^{-1} \sin \alpha_B.$$

(a) Prove that any such pair solves $\phi_{\xi\eta} = \sin\phi$ and $\widetilde{\phi}_{\xi\eta} = \sin\widetilde{\phi}$.

(b) Put $\widetilde{\phi} = 0$ and solve for the branch $0 < \phi < 2\pi$ at finite points. Convert it to physical coordinates and find its velocity.

(c) For density $L = \tfrac{1}{2}(\phi_t^2 - \phi_x^2) - (1 - \cos\phi)$, compute the kink charge $Q = [\phi(+\infty) - \phi(-\infty)]/(2\pi)$, energy and lower-index momentum integral $\int \phi_t \phi_x \mathrm{d}x$. This line solution is not a periodic field for the monodromy theorem.

Hint. Differentiate both first-order equations once across the other variable. For the vacuum partner integrate $\log\tan(\phi/4)$. In the energy integral use the rest-profile derivative $2\operatorname{sech} z$.

Answer. The branch is

$$\phi(t,x) = 4\arctan e^{\gamma(x - v_{\mathrm{tr}} t - x_0)},$$
$$v_{\mathrm{tr}} = \frac{1 - a_B^2}{1 + a_B^2}, \qquad \gamma = \frac{a_B + a_B^{-1}}{2} = \frac{1}{\sqrt{1 - v_{\mathrm{tr}}^2}}.$$

Its charge is one, energy 8γ, and lower-index momentum $-8\gamma v_{\mathrm{tr}}$.

Classical Field Theory — Exercise Sheet 28

Tensor variation, gravitational waves and boundary action

Hints and answers

Use Lecture 28's curvature, inverse-metric and action conventions. All metrics, variations and coordinate maps are smooth. Repeated spacetime indices run from zero to three. Write $\eta = \mathrm{diag}(1,-1,-1,-1)$ only for the flat background. None of the problems asserts global existence for Einstein evolution.

Exercise 28.1 (The full Palatini tensor calculation). Let g_s be a smooth family of Lorentz metrics in fixed coordinates and put

$$k^{\mu\nu} = \partial_s g_s^{\mu\nu}\big|_0, \qquad h_{\mu\nu} = \partial_s (g_s)_{\mu\nu}\big|_0 = -g_{\mu\alpha}g_{\nu\beta}k^{\alpha\beta}, \qquad C^\rho_{\mu\nu} = \delta\Gamma^\rho_{\mu\nu}.$$

Here ∇ is the background Levi–Civita connection.

(a) Prove that C is a tensor of type $(1,2)$ and derive its expression in terms of ∇h. Do not assume that connection coefficients themselves are tensor components.

(b) Vary the full coordinate Riemann formula. Expand both covariant derivatives of C, including all its index terms, to prove $\delta R^\rho{}_{\sigma\mu\nu} = \nabla_\mu C^\rho_{\nu\sigma} - \nabla_\nu C^\rho_{\mu\sigma}$. Contract to obtain the Palatini identity without setting $\Gamma = 0$.

(c) Contract again to derive the inverse-metric boundary vector $v^\mu = \nabla^\mu(g_{\alpha\beta}k^{\alpha\beta}) - \nabla_\nu k^{\mu\nu}$. This is the complete tensor argument promised by Lecture 28.

Hint. The difference of two connections is linear over smooth functions in both vector arguments. Differentiate metric compatibility to obtain C. The two terms from the lower derivative index cancel in the antisymmetrised covariant derivative because the background connection is torsion-free.

Answer. The identities are

$$C^\rho_{\mu\nu} = \frac{1}{2}g^{\rho\lambda}(\nabla_\mu h_{\lambda\nu} + \nabla_\nu h_{\lambda\mu} - \nabla_\lambda h_{\mu\nu}),$$
$$\delta\,\mathrm{Ric}_{\mu\nu} = \nabla_\rho C^\rho_{\nu\mu} - \nabla_\nu C^\rho_{\rho\mu}, \qquad g^{\mu\nu}\delta\,\mathrm{Ric}_{\mu\nu} = \nabla_\mu v^\mu.$$

The connection-product variations are essential away from normal coordinates.

Exercise 28.2 (Vacuum waves, gauge and the two polarisations). Set $g_{\mu\nu} = \eta_{\mu\nu} + s h_{\mu\nu} + O(s^2)$. Raise perturbation indices with η, write $h = \eta^{\mu\nu}h_{\mu\nu}$ and define $\bar h_{\mu\nu} = h_{\mu\nu} - \eta_{\mu\nu}h/2$.

(a) Derive $\Gamma^{(1)}$, $\mathrm{Ric}^{(1)}$ and $G^{(1)}$. Show explicitly that $G^{(1)}_{00}$ and $G^{(1)}_{0i}$ contain no second time derivatives of the perturbation. These are the linearized vacuum constraint equations on a slice.

(b) Under the covariant-metric Lie variation $h_{\mu\nu} \mapsto h_{\mu\nu} + \partial_\mu\xi_\nu + \partial_\nu\xi_\mu$, derive the trace-reversed change and its divergence. In harmonic gauge $\partial^\mu\bar h_{\mu\nu} = 0$, derive the vacuum wave equation. No analytic theorem establishing that gauge globally is assumed or requested.

(c) For a mode $h_{\mu\nu} = \epsilon_{\mu\nu}e^{-i\nu t + i\nu x^3}$ with $\nu > 0$, impose harmonic gauge. Use residual null-wave gauge parameters to set the four temporal amplitudes to zero and derive the complete transverse traceless matrix. Count the independent polarisations. The real field is obtained by taking real parts; the plane mode itself is not claimed to have finite total energy on $\mathbb{R}^3$.

Hint. At first order only derivatives of $\Gamma^{(1)}$ enter the curvature. For the constraints expand the time-index components before imposing harmonic gauge. For the mode use the covector $\ell_\mu = (\nu, 0, 0, -\nu)$ and a parameter $\xi_\mu = \alpha_\mu e^{-i\ell_\rho x^\rho}$.

Answer. In harmonic gauge $G^{(1)}_{\mu\nu} = -\Box\bar{h}_{\mu\nu}/2$. Residual null-wave gauge leaves

$$\epsilon^{\mathrm{TT}}_{\mu\nu} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}.$$

There are two independent transverse traceless polarisations. The constraint components contain spatial and first time derivatives only.

Exercise 28.3 (The non-null boundary action). Let Σ be a smooth non-null boundary patch of a region D. All variations have compact support in a collar of the interior of this patch and vanish near its edges and every other boundary piece. A Gaussian collar for the background may be used as a geometric given:

$$g = \epsilon_n\, \mathrm{d}n^2 + \gamma_{ab}(n, y)\mathrm{d}y^a\mathrm{d}y^b, \qquad \Sigma = \{n = 0\}, \quad D \text{ locally has } n \le 0, \quad \epsilon_n = \pm 1.$$

The outward unit normal is $N = \partial_n$. Write $\mathrm{vol}_\Sigma = \iota_N \mathrm{vol}_g$ with outward-first orientation, $K = \nabla_\mu N^\mu$, and fix only the induced metric: $h_{ab}|_\Sigma = 0$ for $h_{\mu\nu} = \delta g_{\mu\nu}$. Normal components of h are initially unrestricted.

Gaussian normal coordinates for a smooth non-null hypersurface are the local construction described in Wald, *General Relativity*, §3.3, pp. 42–43. The boundary-action discussion is Appendix E.1, pp. 458–459. Use the explicit conventions in this question, including ϵ_n.

(a) Construct a vector field ξ vanishing on Σ such that $\widetilde{h} = h + \mathcal{L}_\xi g$ has $\widetilde{h}_{nn} = \widetilde{h}_{na} = 0$ near the patch. Derive and solve the required first-order normal equations. Explain why the induced variation is still zero.

(b) In this representative compute the normal flux of the Palatini vector and the variation of $2\epsilon_n \int_\Sigma K\mathrm{vol}_\Sigma$. Prove their cancellation.

(c) Use diffeomorphism invariance and Lecture 28's Bianchi identity to extend that result to the original arbitrary normal components. State the resulting gravitational action with the frozen overall sign. Specify why this is not a null-boundary or corner/joint formula.

Hint. The two normal Lie-variation components are $2\epsilon_n\partial_n\xi^n$ and $\gamma_{ab}\partial_n\xi^b + \epsilon_n\partial_a\xi^n$. For a Gaussian-preserving variation with $h_{ab} = 0$ on the boundary, $K = \gamma^{ab}\partial_n\gamma_{ab}/2$. The bulk Einstein pairing is unchanged by a boundary-fixing Lie variation.

Answer. The boundary-corrected gravitational action in this scope is

$$S_{\mathrm{grav}} = -\frac{1}{16\pi G_{\mathrm{N}}}\left[\int_D R_{\mathrm{sc}}\mathrm{vol}_g + 2\epsilon_n\int_\Sigma K\mathrm{vol}_\Sigma\right].$$

The Palatini flux is $-\epsilon_n\gamma^{ab}\partial_n\widetilde{h}_{ab}\mathrm{vol}_\Sigma$, cancelled by the boundary term's variation. No null boundary or joint term is included.